

A 2-D Unstructured Finite-Volume Solver for the Compressible Navier–Stokes Equations: Implementation, MPI Parallelisation, and Benchmark Results

`fvmcns` — submitted solver

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Abstract

This report documents `fvmcns`, a cell-centred unstructured finite-volume solver for the two-dimensional compressible Navier–Stokes equations of a calorically perfect gas, written from scratch in C++17 with mandatory MPI domain decomposition. The solver reads CGNS meshes, partitions the cell dual graph with METIS k -way, and stores only the rank-local partition plus its ghost layer during iterations. The spatial discretisation is second-order: piecewise-linear reconstruction from inverse-distance weighted least-squares gradients, limited with a differentiable Venkatakrishnan limiter and guarded by an explicit positivity fallback. The inviscid flux is HLLC by default, with Roe/Harten–Yee and Rusanov also implemented and selectable at run time. Viscous fluxes use face-corrected primitive-variable gradients with a Newtonian stress tensor and Fourier heat flux. Steady cases are advanced by a CFL-ramped local pseudo-time implicit march; the transient cylinder case uses a true two-level BDF2 dual-time loop with frozen physical-time histories. The inner linear system is solved by block LU-SGS with an analytic flux Jacobian. Of the eight required benchmark cases, 7 reached a converged steady state, 1 reached a statistically periodic state, and 0 are reported as failed. Section 9 gives the per-case status verbatim from each `run_status.json`; no case is presented as successful unless the solver itself recorded it as such, and Section 12 states what did not work, with the evidence: the leading-edge robustness limit of the supersonic inviscid case, the under-predicted Re 200 shedding and its numerical causes, and the cost of the transient inner solve.

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1 Introduction

The benchmark asks for a complete compressible-flow solver rather than a demonstration kernel: mesh import, partitioning, a conservative second-order residual, an implicit nonlinear solve, a physical-time transient path, a machine-readable output contract, and an analysis of eight prescribed cases on two supplied meshes. This report is organised to follow that pipeline, and every quantitative claim is traceable to a file in the submitted result directories.

The eight required cases are three inviscid NACA0012 cases at Mach 0.15, 0.80 and 2.00; three laminar NACA0012 cases at the same Mach numbers with $Re = 5000$; the steady laminar cylinder at $Re = 20$; and the transient vortex-shedding cylinder at $Re = 200$. All NACA cases are at zero angle of attack, which provides a strong and cheap correctness check: lift must vanish by symmetry while drag and the surface pressure distribution must remain non-trivial.

Two design commitments shaped the implementation. First, the solver is *modular by physical concept* rather than by case: gas model, flux, limiter, gradient operator, boundary condition, time integrator and output writer are separate translation units communicating through narrow interfaces, so that a future extension to three dimensions, a general equation of state, a RANS turbulence model or multiple species touches a bounded set of files. Section 11 makes this concrete. Second, *the solver reports what happened*. A run that fails to meet its convergence target is written out with `convergence_status` set to `failed` and a nonzero exit code; the report generator reads that field and propagates it unchanged. This is a deliberate choice because the benchmark explicitly treats a mislabelled failure as worse than an acknowledged one.

2 Governing Equations and Nondimensionalisation

2.1 Conservative form

The solver advances the conservative state vector

$$\mathbf{U} = [\rho, \rho u, \rho v, \rho E]^T, \quad (1)$$

governed by the two-dimensional compressible Navier–Stokes equations

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}^i(\mathbf{U})}{\partial x} + \frac{\partial \mathbf{G}^i(\mathbf{U})}{\partial y} = \frac{\partial \mathbf{F}^v(\mathbf{U}, \nabla \mathbf{U})}{\partial x} + \frac{\partial \mathbf{G}^v(\mathbf{U}, \nabla \mathbf{U})}{\partial y}, \quad (2)$$

with inviscid fluxes

$$\mathbf{F}^i = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ \rho uv \\ u(\rho E + p) \end{bmatrix}, \quad \mathbf{G}^i = \begin{bmatrix} \rho v \\ \rho uv \\ \rho v^2 + p \\ v(\rho E + p) \end{bmatrix}. \quad (3)$$

2.2 Perfect-gas closure

The gas is calorically perfect with configurable γ , R and Prandtl number, all read from the `gas` block of the case file:

$$p = (\gamma - 1)\rho e, \quad E = e + \frac{1}{2}(u^2 + v^2), \quad a = \sqrt{\gamma p / \rho}, \quad T = \frac{p}{\rho R}. \quad (4)$$

All eight benchmark cases use $\gamma = 1.4$, $R = 1$ and $Pr = 0.72$. The conversion between conservative and primitive variables, the sound speed and the temperature are the only places where the equation of state is consulted; they are confined to `src/physics/gas_model.{hpp,cpp}` behind a `GasModel` class, which is the seam a future real-gas or multi-species model would replace.

2.3 Viscous stresses and heat flux

The viscous fluxes use the Newtonian stress tensor with Stokes' hypothesis and a Fourier heat flux,

$$\tau_{xx} = 2\mu u_x - \frac{2}{3}\mu(u_x + v_y), \quad \tau_{yy} = 2\mu v_y - \frac{2}{3}\mu(u_x + v_y), \quad (5)$$

$$\tau_{xy} = \mu(u_y + v_x), \quad \mathbf{q} = -k\nabla T, \quad k = \frac{\mu c_p}{Pr}, \quad (6)$$

with $c_p = \gamma R / (\gamma - 1)$, giving the viscous flux vectors

$$\mathbf{F}^v = \begin{bmatrix} 0 \\ \tau_{xx} \\ \tau_{xy} \\ u\tau_{xx} + v\tau_{xy} - q_x \end{bmatrix}, \quad \mathbf{G}^v = \begin{bmatrix} 0 \\ \tau_{xy} \\ \tau_{yy} \\ u\tau_{xy} + v\tau_{yy} - q_y \end{bmatrix}. \quad (7)$$

Inviscid mode is obtained by setting $\mu = k = 0$, which removes the viscous face loop entirely rather than multiplying by zero.

2.4 Nondimensionalisation and the Reynolds number

The case files are already posed in a consistent nondimensional system: $\rho_\infty = 1$, $|\mathbf{u}_\infty| = 1$, $L_{ref} = 1$, and $p_\infty = 1/(\gamma M_\infty^2)$, which follows from Eq. (4) once $a_\infty = 1/M_\infty$ is imposed. The solver verifies this consistency on input rather than assuming it: the freestream pressure implied by the Mach number is compared against the supplied `freestream.pressure` and a mismatch is a fatal input error. For example the $M = 0.15$ cases supply $p_\infty = 31.746\dots = 1/(1.4 \times 0.15^2)$ and the $M = 0.1$ cylinder cases supply $71.428\dots = 1/(1.4 \times 0.1^2)$.

Laminar cases use a constant viscosity fixed by the case Reynolds number,

$$\mu = \frac{\rho_\infty |\mathbf{u}_\infty| L_{ref}}{Re}, \quad (8)$$

evaluated once at start-up from `physics.reynolds` and `reference.reynolds_length`. This is not a hard-coded constant: for the cylinder at $Re = 20$ the solver records $\mu = 0.05$ in `meta-data.json`, and for the NACA $Re = 5000$ cases $\mu = 2 \times 10^{-4}$, each exactly $1/Re$ as Eq. (8) requires. A Sutherland law is implemented behind the same `GasModel` interface and is selectable through `physics.viscosity_model`, but the benchmark cases all request `constant`.

Force coefficients use the case `reference.area` A_{ref} , `reference.length` and the freestream dynamic pressure $q_\infty = \frac{1}{2}\rho_\infty |\mathbf{u}_\infty|^2$:

$$C_d = \frac{F_x}{q_\infty A_{ref}}, \quad C_l = \frac{F_y}{q_\infty A_{ref}}, \quad C_m = \frac{M_z}{q_\infty A_{ref} L_{ref}}, \quad (9)$$

with the moment taken about `reference.moment_center`. Pressure and viscous contributions are accumulated separately throughout, so the `forces.csv` split is exact rather than reconstructed.

3 Meshes, Boundary Tags, and Case Inputs

3.1 CGNS import

Meshes are read through the CGNS mid-level library in `src/mesh/cgns_reader.cpp`. Both supplied files are ADF-in-HDF5, `CGNSLibraryVersion 3.31`, with `CellDimension = 2` and unstructured zones containing mixed `TRI_3` and `QUAD_4` volume sections plus `BAR_2` boundary sections.

Three properties of these particular files required care and are worth recording, because a reader that assumes the common case fails silently on them:

1. **The cylinder mesh has two zones joined by `GridConnectivity_t` nodes, not `GridConnectivity1to1_t`.** `cg_n1to1` returns zero, so the interfaces are reachable only through `cg_nconns/cg_conn_read`. The reader merges the two zones by fusing the matched vertex pairs, reducing $7079 + 3236 = 10315$ raw vertices to 9995 unique vertices. A reader that only queried `cg_n1to1` would produce two disconnected meshes.
2. **Boundary conditions carry `BCType = FamilySpecified` with `cg_fambc_read` returning `BCTypeNull`.** The physical type is therefore *not* recoverable from CGNS enums; it must come from the family *name*. The solver maps family names to physics through the case file's `boundary_conditions` block, so the NACA tags `bc-2` and `bc-4` (which are semantically empty) and the cylinder tags `WALL` and `FAR` are all handled by the same mechanism. An unmapped or unknown family is a fatal input error, as required by the CLI contract.
3. **The NACA base declares `PhysicalDimension = 3` and carries a vestigial `CoordinateZ` array** that is not identically zero (15198 of 15682 values are nonzero, maximum magnitude 3.46×10^{-6}). The reader drives dimensionality from `CellDimension` and ignores the third coordinate.

3.2 Geometry construction and verification

From the vertex coordinates and element connectivity the solver builds cell centroids and areas by the polygon shoelace formula, then constructs the face list by hashing sorted vertex pairs: a pair seen twice becomes an interior face with a left and right cell, a pair seen once becomes a boundary face whose tag is inherited from the `BAR_2` section containing it. Face normals are oriented outward from the left cell.

The resulting geometry was cross-validated against an independent Python CGNS probe (`tools/probe_mesh.py`) written without reference to the C++ code. Table 1 lists the agreed values. Two global identities are checked at start-up on every run: the sum of signed cell areas must equal the total domain area, and the sum of outward face-normal vectors around every cell must vanish. The latter closes to $\sim 10^{-15}$ on both meshes, which confirms consistent normal orientation — the single most common source of silent conservation errors in an unstructured code.

Table 1: Verified mesh properties. Values agreed between the C++ reader and an independent Python CGNS probe (`tools/probe_mesh.py`); see also `docs/MESH.NOTES.md`.

mesh	zones	vertices	cells	tri/quad	interior f.	bdry f.	families	extent
NACA0012.H2.cgns	1	15682	20816	10752 / 10064	36014	484	bc-2 far field, bc-4 wall	± 80
CylinderB1.cgns	2 (merged)	9995	10185	500 / 9685	20060	120	WALL wall, FAR far field	± 200

Table 2 completes the picture with the flow parameters of every case. It is generated from the benchmark case JSON, so it states the inputs actually consumed rather than a transcription of them; the viscosity column is the derived $\mu = \rho_\infty U_\infty L_{ref}/Re$ that the solver computes at start-up, and it agrees with the `solver_details.constant_viscosity` recorded in each `metadata.json`.

Table 2: Flow parameters for all eight required cases, read from `inputs/cases/*.json`. Inviscid cases have no Reynolds number and no viscosity. μ is derived, not supplied.

case	mesh	mode	M_∞	AoA	Re	μ	γ	run
naca0012_m015_inviscid	NACA0012.H2	inviscid	0.15	0.0	—	—	1.4	steady
naca0012_m080_inviscid	NACA0012.H2	inviscid	0.80	0.0	—	—	1.4	steady
naca0012_m200_inviscid	NACA0012.H2	inviscid	2.00	0.0	—	—	1.4	steady
naca0012_m015_laminar_re5000	NACA0012.H2	laminar	0.15	0.0	5000	0.0002	1.4	steady
naca0012_m080_laminar_re5000	NACA0012.H2	laminar	0.80	0.0	5000	0.0002	1.4	steady
naca0012_m200_laminar_re5000	NACA0012.H2	laminar	2.00	0.0	5000	0.0002	1.4	steady
cylinder_m010_laminar_re20	CylinderB1	laminar	0.10	0.0	20	0.05	1.4	steady
cylinder_m010_laminar_re200	CylinderB1	laminar	0.10	0.0	200	0.005	1.4	transient

The NACA mesh contains extremely small cells at the leading and trailing edge: the smallest cell areas are of order 5×10^{-9} against a domain area of 2.0085×10^4 , a span of roughly twelve orders of magnitude. This range is the origin of two numerical difficulties discussed later — the choice of residual norm (Section 6.1) and the convergence stall of the shocked NACA cases (Section 12).

3.3 Case inputs

Case files are JSON, parsed with `nlohmann/json` in `src/core/case_input.cpp`. Parsing is *checked*: unknown required keys, non-positive Reynolds numbers, inconsistent freestream pressure, un-mapped boundary families and out-of-range CFL values are all fatal with a specific message. No case-specific branch exists anywhere in the solver; the eight cases differ only in their input files and, where noted, in documented command-line options. Table 6 lists the run controls actually used.

4 Spatial Discretisation

4.1 Finite-volume residual

Integrating Eq. (2) over cell i of area V_i and applying the divergence theorem gives the semi-discrete form

$$V_i \frac{d\mathbf{U}_i}{dt} = -\mathbf{R}_i, \quad \mathbf{R}_i = \sum_{f \in \partial i} [\mathcal{F}^i(\mathbf{U}_L, \mathbf{U}_R, \mathbf{n}_f) - \mathcal{F}^v(\mathbf{U}, \nabla \mathbf{U}, \mathbf{n}_f)] A_f, \quad (10)$$

where A_f is the face length, $\mathbf{n}_f$ the outward unit normal, and $\mathcal{F}^i$, $\mathcal{F}^v$ the numerical inviscid and viscous fluxes. The implementation in `src/numerics/residual.cpp` loops faces once and scatters $\pm$ the same flux vector to the two adjacent cells, so the discretisation is conservative by construction: an interior face can never contribute unequal amounts to its two neighbours.

4.2 Second-order reconstruction

Cell gradients of the primitive variables $\mathbf{w} = [\rho, u, v, p]^T$ are computed by inverse-distance weighted least squares over the face-neighbour stencil. For cell i the gradient minimises

$$\sum_{j \in N(i)} w_j (\nabla \mathbf{w}_i \cdot \mathbf{d}_j - \Delta \mathbf{w}_j)^2, \quad w_j = \frac{1}{|\mathbf{d}_j|^2}, \quad \mathbf{d}_j = \mathbf{x}_j - \mathbf{x}_i, \quad (11)$$

whose 2×2 normal equations are inverted analytically. Because the geometric weights and the stencil are fixed, the inverse matrix is folded into per-neighbour coefficient vectors *once* at set-up (`GradientOperator::build_least_squares`); each subsequent gradient evaluation is then a single sparse pass with no linear algebra. Degenerate stencils — a cell whose neighbours are nearly collinear, which does occur at the mesh boundary — are regularised by an additive term scaled by the matrix magnitude, so the regularisation is dimensionally consistent and vanishes for well-posed stencils. A Green–Gauss operator is implemented behind the same interface and selectable via `scheme.gradient`.

Face states follow from limited piecewise-linear reconstruction,

$$\mathbf{w}_L = \mathbf{w}_i + \Phi_i \nabla \mathbf{w}_i \cdot (\mathbf{x}_f - \mathbf{x}_i), \quad (12)$$

with $\Phi_i \in [0, 1]$ the per-variable limiter factor. Reconstruction is active in *all* production runs reported here; the only first-order paths are the supersonic start-up ramp described in Section 6, which is switched off well before convergence, and the per-cell positivity fallback below. Both are counted and reported, and the counts are given in Table 3 rather than asserted.

The honest reading is that the fallback is small but *not* everywhere zero. Two cases — the transonic inviscid airfoil and the $Re = 20$ cylinder — end with no first-order cells at all. The two Mach 2 cases end with 2.55% and 2.67% of cells first order, and the remaining three with under 0.1%. The affected cells are the microscopic leading-edge cells discussed at length in Section 12.1, which is where a supersonic stagnation point meets the smallest cells in the mesh; that is exactly where a positivity fallback is supposed to act. No case is a disguised first-order solution, but neither is any shocked case purely second order at the leading edge, and the earlier version of this report overstated the position by claiming a uniform zero.

Table 3: First-order and positivity-limited cell counts in the final step of each completed case, read from `metadata.json`. The fraction is of `num_cells_global`.

case	cells	first-order cells	fraction	positivity-limited
naca0012_m015_inviscid	20816	12	0.06%	0
naca0012_m080_inviscid	20816	0	0.00%	0
naca0012_m200_inviscid	20816	531	2.55%	0
naca0012_m015_laminar_re5000	20816	18	0.09%	0
naca0012_m080_laminar_re5000	20816	4	0.02%	0
naca0012_m200_laminar_re5000	20816	556	2.67%	0
cylinder_m010_laminar_re20	10185	0	0.00%	0
cylinder_m010_laminar_re200	10185	0	0.00%	0

4.3 Limiter and positivity control

The default limiter is the differentiable Venkatakrishnan form of the Barth–Jespersen limiter. With $\Delta_{\max}$, $\Delta_{\min}$ the extreme neighbour deviations from $\mathbf{w}_i$ and Δ the unlimited reconstruction incre-

ment towards a face,

$$\Phi = \frac{(y^2 + 2y) + \varepsilon}{(y^2 + y + 2) + \varepsilon}, \quad y = \frac{\Delta_{\pm}}{\Delta}, \quad \varepsilon = (K h_i)^3, \quad (13)$$

so that Φ is dimensionless and lies in $[0, 1]$, taking $\Delta_+ = \Delta_{\max}$ when $\Delta > 0$ and $\Delta_- = \Delta_{\min}$ otherwise, with h_i the local length scale and $K = 5$ by default. Two branches precede this expression and are part of the definition: if the increment towards a face is positive while $\Delta_{\max} \leq 0$, or negative while $\Delta_{\min} \geq 0$, then Φ is set to zero directly. Those are the conditions at a local maximum and a local minimum respectively, so a cell sitting at a smooth extremum is driven to first order regardless of ε . That is the classical extremum-clipping weakness of Barth–Jespersen-type limiters, it is measured for the Re 200 wake in Section 12.3, and it is the reason the ε threshold alone does not make this limiter benign in smooth flow.

The ε threshold is what makes the limiter differentiable: it disables limiting in smooth low-amplitude regions, which preserves second-order accuracy and — more importantly for an implicit solver — prevents the limiter from switching state between iterations and stalling convergence. The stricter, non-differentiable Barth–Jespersen limiter is also available via `scheme.limiter`.

4.3.1 Limiter freezing, and why it is necessary here

The ε threshold above is *dimensionally inert on these meshes*, and this turned out to be the single most consequential numerical finding of the study. Because $\varepsilon = (K h_i)^3$ scales as the cube of the cell length, the leading-edge cells with $h_i \approx 9.2 \times 10^{-7}$ give $\varepsilon \approx 9.7 \times 10^{-17}$, against an actual variation scale $\Delta^2 \approx 10^{-4}$ — twelve orders of magnitude too small. The smoothing term therefore never activates where it is needed, and the “differentiable” limiter degenerates into hard min / max switching, including branches that set Φ exactly to zero.

The measured consequence is that between 2021 and 10264 cells changed their limiter factor by more than 1% *on every step*, with a maximum change of exactly 1.0. The residual operator $\mathbf{R}(\mathbf{U})$ is thus not differentiable, and a Newton-type method cannot converge against it: it can only limit-cycle. On the Mach 2 inviscid case the resulting cycle is period-20, confirmed by an autocorrelation of 0.877 at lag 20.

The remedy used for production is to *freeze* the limiter factors once the residual has fallen far enough that the limiting pattern is established. After freezing, the remaining iteration is a smooth Newton solve on a fixed reconstruction stencil. Gradients remain active, and the positivity enforcement of the previous section still runs at every evaluation, so nothing about the reconstruction’s safety is weakened. Freezing is triggered only when all of the following hold: second-order reconstruction is active, the CFL ramp is complete, and the residual reduction *averaged over a rolling window* of `limiter_freeze_hold_steps` exceeds `limiter_freeze_orders`. The event is reported in the run notes.

The rolling window matters, and the reason is instructive. An earlier version required a run of *consecutive* qualifying steps, which can never trigger on precisely the cases that need it: the pre-freeze oscillation is itself what breaks the run. On the Mach 2 case, 69% of post-ramp steps cleared the threshold but the longest unbroken run was only 55 steps against a 200-step requirement, so the gate never fired. A windowed mean is insensitive to that chatter while still demanding a sustained improvement.

Freezing is what rescues the transonic case: `naca0012_m080_inviscid` previously exhausted all 30 000 permitted steps at 2.95 orders, and now converges in 3250 steps to 4.43 orders. The drag changes only in its sixth decimal across the freeze event, so the technique is not biasing the result.

This steady freeze never triggers on the subsonic or viscous steady cases, which converge before their CFL ramp completes.

A second, separate freeze applies inside the transient inner loop. The mechanism above is a *steady-path* device gated on the CFL ramp and a windowed residual history. The transient driver carries a distinct one, and it must be described separately because it behaves quite differently: it freezes the limiter within a physical step once that step’s own inner-residual ratio falls to `transient_limiter_freeze_ratio = 0.5`, with no CFL gate, and it is enabled by default. It therefore fires on essentially every one of the 30 000 Re 200 physical steps, and it is the change responsible for the reduction in mean inner iterations reported in Section 12.4.

Two properties keep it sound. The exact limiter is used for the early, large-amplitude inner iterations, so the freeze only takes effect once that step’s iterate has already settled; and the freeze is *released and re-derived at every physical step*, so a frozen pattern can never persist into a state it no longer describes — which is the obvious objection, and the failure mode actually observed on the supersonic steady case. Unlike the steady freeze this one is logged only under `--profile`, so it is not visible in the default run notes; that is a diagnostics gap rather than a behavioural one, and the setting is recorded in `metadata.json`.

One honest caveat is recorded in Section 12: freezing is effective when the update clip is dormant, but it can slightly *widen* the residual oscillation when the clip is actively firing, because the held limiter pattern then drifts out of correspondence with the clip-modified state.

Positivity is protected at two distinct points, because limiting alone is not sufficient:

1. **At reconstructed face states.** After limiting, every reconstructed state is tested for positive density and pressure. Where a face state is inadmissible the cell’s slope is halved — at most twelve times, after which the slope is $2^{-12} \approx 2 \times 10^{-4}$ of the original — until every face state satisfies both admissibility and the stricter floors $\rho_f > 0.1\rho_i$, $p_f > 0.1p_i$; if the reduced slope is still inadmissible the cell is set genuinely first order. The number of cells affected is reported as `positivity_limited_cells_final_step`.
2. **At the state update.** After each implicit step the update is applied per cell with a factor $f \leq 1$ chosen so that the new state stays physical and changes ρ and p by no more than a bounded fraction (`apply_update_with_positivity_clip`). Bounding the *fractional* change, rather than merely the sign, prevents a strong start-up transient from driving a cell to a positive but physically meaningless pressure that would then poison its neighbours’ stencils.

A single global relaxation factor is deliberately *not* used for step control. With cell areas spanning twelve orders of magnitude, one 10^{-9} -area trailing-edge cell would otherwise dictate the step for twenty thousand healthy cells. The global factor is reserved for genuine failure: if any increment anywhere is non-finite the step is rejected outright and the Jacobian is rebuilt.

4.4 Inviscid flux

Three approximate Riemann solvers are implemented in `src/physics/riemann_flux.cpp` and selected by `scheme.inviscid_flux` or `--flux`:

- **HLLC** (default, `hllc`). Wave speeds from the Davis estimates with a Roe-averaged intermediate, a contact speed from the momentum balance, and the standard three-state reconstruction. HLLC is the production choice because it resolves the contact discontinuity — which matters for the viscous wake and boundary layer — while remaining positivity friendly at the leading-edge shock.

- **Roe with a Harten–Yee entropy fix.** The classical flux-difference splitting over the Roe average, with each characteristic wave speed floored by $|\lambda| \leftarrow (\lambda^2 + \delta^2)/(2\delta)$ for $|\lambda| < \delta$. The fix suppresses the expansion shock that an unmodified Roe flux admits at a sonic point.
- **Rusanov / local Lax–Friedrichs.** The most dissipative option, retained as a robustness fallback and because the Re 200 case file specifies a Rusanov dissipation scale.

4.5 Viscous flux

Viscous fluxes need a face gradient, and the naive arithmetic average of the two cell gradients is inconsistent on stretched cells — exactly the cells that resolve a boundary layer. The solver therefore uses the standard face-corrected average: for each primitive variable,

$$\nabla \mathbf{w}_f = \overline{\nabla \mathbf{w}} + \frac{(\mathbf{w}_R - \mathbf{w}_L) - \overline{\nabla \mathbf{w}} \cdot \mathbf{d}_{LR}}{|\mathbf{d}_{LR}|^2} \mathbf{d}_{LR}, \quad \overline{\nabla \mathbf{w}} = \frac{1}{2} (\nabla \mathbf{w}_L + \nabla \mathbf{w}_R), \quad (14)$$

which reproduces the two-point difference along the cell-to-cell direction while retaining the tangential slope from the averaged gradient. The temperature gradient is obtained from the density and pressure gradients via Eq. (4),

$$\nabla T = \frac{\nabla p - TR \nabla \rho}{\rho R}, \quad (15)$$

so that no separate temperature field has to be stored or exchanged.

At a wall the boundary state supplies the face state, and the wall-normal gradient component is replaced by the one-sided difference between the cell centre and the boundary value over the wall distance — the consistent one-sided form on a stretched mesh. For an adiabatic wall the normal component of ∇T is then projected out exactly, so the wall heat flux vanishes identically rather than approximately.

5 Boundary Conditions

All boundary conditions are implemented in `src/physics/boundary_conditions.cpp` as a function returning *two* states for a boundary face: an `exterior` state fed to the Riemann solver, and a `boundary` state representing the physical value *on* the wall. Keeping these distinct is what allows the surface output to report true boundary values rather than adjacent cell-centre values.

Far field. A characteristic-based treatment. The local normal Mach number and flow direction select between subsonic and supersonic, inflow and outflow; for subsonic boundaries the incoming characteristics carry freestream information and the outgoing ones carry interior information, combined through the Riemann invariants. This makes the far field weakly reflecting, which matters because the NACA far field is only 80 chords away.

Inviscid slip wall. The exterior state mirrors the normal velocity component, $\mathbf{u}_{ext} = \mathbf{u}_{int} - 2(\mathbf{u}_{int} \cdot \mathbf{n})\mathbf{n}$, leaving density and pressure unchanged. The Riemann solver applied to this mirrored pair yields a flux with zero mass transfer and pressure-only momentum transfer, which is the correct slip-wall condition, and it does so without special-casing the flux function. The boundary state has its normal component projected out, preserving the tangential velocity.

Viscous no-slip adiabatic wall. The boundary state has $\mathbf{u} = \mathbf{0}$ exactly, with density and pressure taken from the interior cell, which is the standard zero normal-pressure-gradient approximation. The exterior state reverses the full velocity vector so that the arithmetic mean seen by the flux function is zero velocity. The adiabatic condition is enforced by projecting the normal temperature gradient to zero as described above, so no energy crosses the wall.

Surface output semantics. `surface.csv` reports the boundary state, recorded in `meta-data.json` as `wall_boundary_output_semantics = boundary_value`. This is verified numerically in Section 8: no-slip rows carry velocity identically zero, and inviscid slip-wall rows carry $|\mathbf{u} \cdot \mathbf{n}| \sim 10^{-11}$ with tangential velocity retained.

6 Implicit and Transient Time Integration

6.1 Residual norm

The reported residual norm deserves an explicit definition because on these meshes the choice is not cosmetic. The raw residual $\mathbf{R}_i$ of Eq. (10) has units of flux $\times$ area and therefore scales with cell size. On a mesh whose cell areas span twelve orders of magnitude, a raw norm is dominated by the largest far-field cells and can *rise* simply because a transient propagated outward. The solver therefore reports the volume-normalised residual $\mathbf{R}_i/V_i$, which is precisely the local rate of change $d\mathbf{U}_i/dt$ — a mesh-size-independent measure of “the solution has stopped changing”:

$$\|R\|_2 = \sqrt{\frac{1}{N} \sum_{i=1}^N \sum_{k=1}^4 \left(\frac{R_{i,k}}{V_i} \right)^2}, \quad \|R\|_\infty = \max_{i,k} \left| \frac{R_{i,k}}{V_i} \right|. \quad (16)$$

Both the sums and the maxima are reduced across all ranks with MPI collectives, so every reported norm is a true global value and never a rank-local one. Convergence is measured as orders of magnitude of reduction from the first step, $\log_{10}(\|R\|_2^{(1)}/\|R\|_2^{(n)})$, against the per-case `residual-reduction-target`.

The trade-off of this norm is that it weights a microscopic cell exactly as heavily as a large one. Section 12 shows that this is the mechanism behind the slow terminal convergence of the shocked NACA cases, where a handful of 5×10^{-9} -area leading-edge cells dominate $\|R\|_\infty$ long after the forces have become stationary to four significant figures.

A volume-weighted RMS, $\sqrt{\sum_i V_i (R_i/V_i)^2 / \sum_i V_i}$, was implemented and instrumented alongside the norm above as a candidate replacement, because 99.4% of the unweighted norm was traced to 592 cells of area below 10^{-8} . It was *rejected*: measured on the same residual fields, the two norms plateau together (unweighted 2.0–2.2 orders, volume-weighted 1.85–2.26). Switching would therefore have relabelled the reported number without converging anything, while obscuring a real defect. The underlying error in those cells was genuine and is resolved by the fixes of Section 4.3.1 rather than by a change of measure.

6.2 Steady path: CFL-ramped local pseudo-time

6.2.1 Convergence criterion

A steady run is declared converged only when *both* of the following hold: the residual has fallen by the case’s `residual-reduction-target` orders, *and* the drag coefficient has stopped changing. The second condition is not redundant, and omitting it produced a false positive that is worth recording.

The reference residual is measured on the first-order start-up state, which for a supersonic case is large enough that the orders target can be crossed on a transient dip while the solution is still far from steady. Under an orders-only test, `naca0012_m200_laminar_re5000` was reported converged with $C_d = 3.098$ while C_d was still falling at 18% per 50 steps — some thirty times the physically plausible value. The run was numerically well-behaved; only the stopping rule was wrong.

The settling test is deliberately absolute-*or*-relative: C_d is settled if either $|\Delta C_d| \leq 10^{-8}$ or $|\Delta C_d/C_d| \leq 10^{-3}$ over a 50-step separation. The absolute arm is load-bearing rather than cosmetic. The exact inviscid drag of a symmetric airfoil at zero incidence is *zero*, so a purely relative test is ill-posed for `naca0012_m015_inviscid` — the denominator tends to zero and last-digit noise is amplified into apparent drift. Only C_d is tested; C_l is numerically fragile at zero incidence ($\mathcal{O}(10^{-5})$) and is deliberately excluded.

A monotonicity criterion (reject if recent samples all move one way) was also tried and *rejected* after measurement: a converging steady solve normally approaches its drag smoothly from one side, and both `cylinder_m010_laminar_re20` and `naca0012_m015_laminar_re5000` — whose drag values are independently corroborated — show a fully monotone approach. The test would have produced false negatives on known-good cases. The rejection is documented in the source so it is not reintroduced.

The practical effect of the settling requirement is that three cases now run longer and report better-converged forces. Most consequentially, `naca0012_m015_laminar_re5000` moves from $C_d = 0.0710$ at 1942 steps to $C_d = 0.05431$ at 3400 steps. The converged value is the defensible one: its viscous component of 0.0353 compares well with the Blasius flat-plate estimate $2 \times 1.328/\sqrt{Re} = 0.0376$ scaled to the airfoil’s wetted length, whereas 0.0710 admits no such interpretation. Every case still converges well inside its permitted `max_steps`: six of the seven steady cases use 5–11% of the allowance and the transonic laminar case, the slowest, uses 17%. The stricter test therefore costs budget headroom but does not threaten it.

Steady cases march the pseudo-time problem

$$\left[\frac{V_i}{\Delta\tau_i} \mathbf{I} + \frac{\partial \mathbf{R}}{\partial \mathbf{U}} \right] \Delta \mathbf{U} = -\mathbf{R}(\mathbf{U}^n), \quad \mathbf{U}^{n+1} = \mathbf{U}^n + f \Delta \mathbf{U}, \quad (17)$$

with a *local* pseudo time step from the cell spectral radius,

$$\Delta\tau_i = \frac{\text{CFL} \cdot V_i}{\sum_{f \in \partial i} (|\mathbf{u} \cdot \mathbf{n}| + a + 2\nu_f) A_f}, \quad \nu_f = \max\left(\frac{2\mu}{\rho \delta n}, \frac{\gamma\mu}{\rho Pr \delta n} \right), \quad (18)$$

so that both the convective and the viscous spectral radii are accounted for, the latter with the momentum and thermal diffusivities. The CFL number follows a geometric ramp from `cfl_initial` to `cfl_max` over `pseudo_cfl_ramp_steps`, which is gentler at start-up than a linear ramp and reaches the cap exactly at the end of the window. The supersonic cases additionally begin with a short first-order start-up phase to get past the initial shock formation, after which second-order reconstruction is enabled for the remainder of the run; the transition step is logged, and as noted above the final state has zero first-order cells.

6.3 Run-control parameters actually used

Table 4 lists the run-control parameters for every case. Both columns are generated rather than typed: the supplied values are read from the benchmark case JSON and the used values from each run’s `metadata.json`, so the table cannot drift from either the inputs or the results. Where the two differ the cell shows *supplied* $\rightarrow$ *used* with a dagger.

Exactly one entry is flagged, `naca0012_m200_inviscid` at $\text{CFL}_{\max} = 25$ rather than the supplied 50, which is the deviation justified in Section 12.1. Every other case ran at its supplied values, and in particular the Re 200 transient ran at the required $\Delta t = 0.01$, $t_{\text{final}} = 300$, inner bounds 5–1000 and inner target 10^{-3} . The Re 200 row has no `max_steps` because the transient path is bounded by `final_time` instead.

Two further command-line options are used for the Re 200 case, `--lusgs-sweeps 2` and `--linear-check-interval 3`. Neither is a benchmark parameter: the first states the default sweep count explicitly and the second checks the linear residual every third inner iteration rather than every one, which matters only because the norm reduction dominates inner-iteration cost under MPI oversubscription (Section 12.4). Both are recorded in `scripts/run_production.sh` and `report/run_manifest.csv`.

Table 4: Run-control parameters per case. Supplied values come from the benchmark case files, used values from the submitted `metadata.json`. A dagger marks the single documented deviation.

case	max_steps	CFL ₀	CFL _{max}	ramp	inner min	inner max	inner target
<code>naca0012_m015_inviscid</code>	20000	1	100	2000	3	50	0.01
<code>naca0012_m080_inviscid</code>	30000	1	100	3000	3	50	0.01
<code>naca0012_m200_inviscid</code>	40000	0.2	50 → 25 [†]	5000	3	80	0.01
<code>naca0012_m015_laminar_re5000</code>	40000	1	100	5000	3	80	0.01
<code>naca0012_m080_laminar_re5000</code>	40000	0.5	80	5000	3	80	0.01
<code>naca0012_m200_laminar_re5000</code>	50000	0.2	50	7000	3	100	0.01
<code>cylinder_m010_laminar_re20</code>	30000	1	100	3000	3	80	0.01
<code>cylinder_m010_laminar_re200</code>	10000	1	1	0	5	1000	0.001

6.4 Transient path: true two-level BDF2 dual time

The Re 200 cylinder uses BDF2 in physical time. Writing $\mathbf{U}^{n+1}$ for the state being solved and freezing the histories, the nonlinear system at each physical step is

$$\mathbf{R}^{\text{tot}}(\mathbf{U}^{n+1}) = \mathbf{R}(\mathbf{U}^{n+1}) + V_i \frac{a_0 \mathbf{U}^{n+1} - a_1 \mathbf{U}^n + a_2 \mathbf{U}^{n-1}}{\Delta t} = \mathbf{0}, \quad (19)$$

with $(a_0, a_1, a_2) = (3/2, 2, 1/2)$ for BDF2. The first physical step uses BDF1, $(1, 1, 0)$, because $\mathbf{U}^{n-1}$ does not yet exist; from the second step onward the scheme is second-order in time.

The loop structure is genuinely two-level, as the benchmark requires:

<pre> for $n = 1 \dots N$ $\mathbf{U}^{n+1} \leftarrow$ initial guess; freeze $\mathbf{U}^n, \mathbf{U}^{n-1}$ for $m = 1 \dots m_{\max}$ evaluate $\mathbf{R}^{\text{tot}}$ from Eq. (19) solve $[\frac{V}{\Delta \tau} + \frac{a_0 V}{\Delta t} + \frac{\partial \mathbf{R}}{\partial \mathbf{U}}] \Delta \mathbf{U} = -\mathbf{R}^{\text{tot}}$ by block LU-SGS $\mathbf{U}^{n+1} \leftarrow \mathbf{U}^{n+1} + \Delta \mathbf{U}$ stop when $\ \mathbf{R}^{\text{tot}}\ / \ \mathbf{R}^{\text{tot},(1)}\ \leq \varepsilon_{\text{inner}}$ and $m \geq m_{\min}$ accept $\mathbf{U}^{n+1}$; only now shift $\mathbf{U}^{n-1} \leftarrow \mathbf{U}^n, \mathbf{U}^n \leftarrow \mathbf{U}^{n+1}$ </pre>	(outer physical time step) (inner nonlinear iterations)
---	---

Three contract points are satisfied explicitly and are worth stating directly. First, $\mathbf{U}^n$ and $\mathbf{U}^{n-1}$ are held in separate arrays and are *never* written during the inner loop; the shift happens after the step is accepted, recorded in metadata as `bdf2_history_update = after_inner_convergence`. Second, the inner convergence target is measured on the *full* transient residual $\mathbf{R}^{\text{tot}}$ of Eq. (19)

including the physical-time term, not on the spatial residual alone. Third, forces and residuals are sampled per accepted physical step, so `forces.csv` is a physical-time history rather than a pseudo-time trace, and `true_bdf2_inner_loop` is recorded `true` in metadata.

6.5 Inner linear solve: block LU-SGS

The linear system in Eqs. (17) and (19) is solved by symmetric block Gauss–Seidel (LU-SGS) on 4×4 blocks, in `src/solve/implicit_solver.cpp`. The flux Jacobian is assembled analytically with a Rusanov-style splitting,

$$\frac{\partial \mathcal{F}}{\partial \mathbf{U}_L} = \frac{1}{2} (\mathbf{A}_L + \lambda \mathbf{I}), \quad \frac{\partial \mathcal{F}}{\partial \mathbf{U}_R} = \frac{1}{2} (\mathbf{A}_R - \lambda \mathbf{I}), \quad \lambda = \sigma \max_{L,R} (|\mathbf{u} \cdot \mathbf{n}| + a) \frac{A_f}{2}, \quad (20)$$

where $\mathbf{A} = \partial \mathbf{F}^i / \partial \mathbf{U}$ is the exact analytic inviscid flux Jacobian and σ a tunable dissipation scale (`--jacobian-dissipation`). Linearising with the Rusanov speed rather than the exact HLLC Jacobian makes the operator strongly diagonally dominant, which is what LU-SGS needs to converge; the second-order HLLC residual still determines the converged solution, so this affects the *rate* of the inner solve and not the answer. Viscous coupling enters as a spectral contribution to the diagonal magnitude.

Each iteration performs a forward sweep over increasing cell index and a backward sweep over decreasing index, using the most recent neighbour increments, then exchanges ghost increments. Critically, the loop then computes the *true* linear residual $\|\mathbf{b} - \mathbf{A} \Delta \mathbf{U}\|$ with a global reduction and iterates until a requested tolerance is met. This makes it a genuine iterative linear solve with a measurable convergence history, not a single diagonal update: the reported per-step linear residual ratio is what the solver actually achieved.

7 MPI Parallelisation and METIS Partitioning

7.1 Partitioning

Partitioning uses METIS *k*-way on the *cell dual graph*: one graph vertex per mesh cell, one graph edge per interior mesh face. Preprocessing is serial on rank 0 — the full mesh is read once, the dual graph built, and `METIS_PartGraphKway` called — which the benchmark explicitly permits. The resulting partition vector is broadcast, and every rank then constructs *only* its own rank-local mesh. Metadata records `partitioner = metis_kway`.

7.2 Rank-local mesh and ghost construction

Each rank stores its owned cells followed by a ghost layer containing exactly the off-rank cells referenced by its faces — one ghost layer, because the stencil (face neighbours for gradients, face pairs for fluxes) reaches exactly one cell deep. Local indices are assigned so that owned cells occupy $[0, N_{\text{own}})$ and ghosts $[N_{\text{own}}, N_{\text{total}})$, which makes the ubiquitous “is this cell mine” test a single integer comparison and lets the residual loop scatter to owned cells without a branch on ownership.

The claim that iterations do not replicate global data is structural, not incidental: after set-up no rank holds an array dimensioned by the global cell count. `metadata.json` records `full_mesh_replication_during_iterations = false` and `full_state_replication_during_iterations = false`, and the per-rank owned/ghost counts in Table 5 sum to appreciably more than the global cell count only by the (small) ghost overlap — $\sum_r N_{\text{own}}^{(r)}$ equals the global count exactly.

7.3 Halo exchange

Communication is neighbour-scoped non-blocking point-to-point (`neighbor_isend_irecv`): each rank posts `MPI_Irecv` for every neighbour, packs and posts `MPI_Isend`, then waits. Send and receive lists are built once at set-up and reused, so no index exchange happens per iteration. No `MPI_Allgather` of state occurs anywhere in the iteration loop; the only collectives are the scalar reductions for residual norms, force sums and the linear-solver residual.

Three quantities are exchanged per residual evaluation, and all three are necessary for parallel–serial equivalence:

1. **Conservative states**, before primitive conversion.
2. **Primitive gradients**, after the gradient pass — so a ghost cell carries the gradient its *owner* computed. Without this a partition-boundary face would reconstruct from a zero or stale slope on its right side and the parallel result would differ from serial.
3. **Limiter factors**, so the reconstruction from the ghost side matches exactly what the owning rank would produce.

Omitting the second or third exchange produces a solver that still converges but to a slightly rank-count-dependent answer — a defect that is invisible without the rank-count comparison in Section 10.

Table 5: Per-rank partition diagnostics at $np = 8$ for `naca0012.m015.inviscid`, from `partition_diagnostics.csv`. Edge cut 431; owned cells min/mean/max 2460/2602.0/3076; load-balance ratio (max/mean) 1.1822. Partitioner `metis_kway`.

rank	owned cells	ghost cells	neighbour ranks	send cells	recv cells
0	2504	81	2 (1 2)	80	81
1	2534	127	4 (0 2 6 7)	128	127
2	2563	90	4 (0 1 3 6)	90	90
3	2460	103	4 (2 5 6 7)	103	103
4	2536	104	2 (5 7)	103	104
5	2516	99	3 (3 4 7)	100	99
6	3076	68	4 (1 2 3 7)	68	68
7	2627	172	5 (1 3 4 5 6)	172	172

The ghost layer is small relative to the owned partition — of order 3–7% of owned cells — which is the expected surface-to-volume ratio for a two-dimensional k -way partition and confirms that the communication volume is a boundary effect rather than a bulk cost. Neighbour counts of 2–5 per rank show METIS produced compact, well-connected subdomains.

8 Output, Reproducibility, and Sanity Checks

8.1 Build and run

Dependencies are MPI, CGNS, HDF5, METIS and header-only `nlohmann/json`; Eigen is detected but optional. Locations are configurable through `CFD_EXTERNALS_ROOT` and `CFD_EXTERNALS_DIR` or the matching CMake cache variables, so the build does not assume a fixed absolute path:

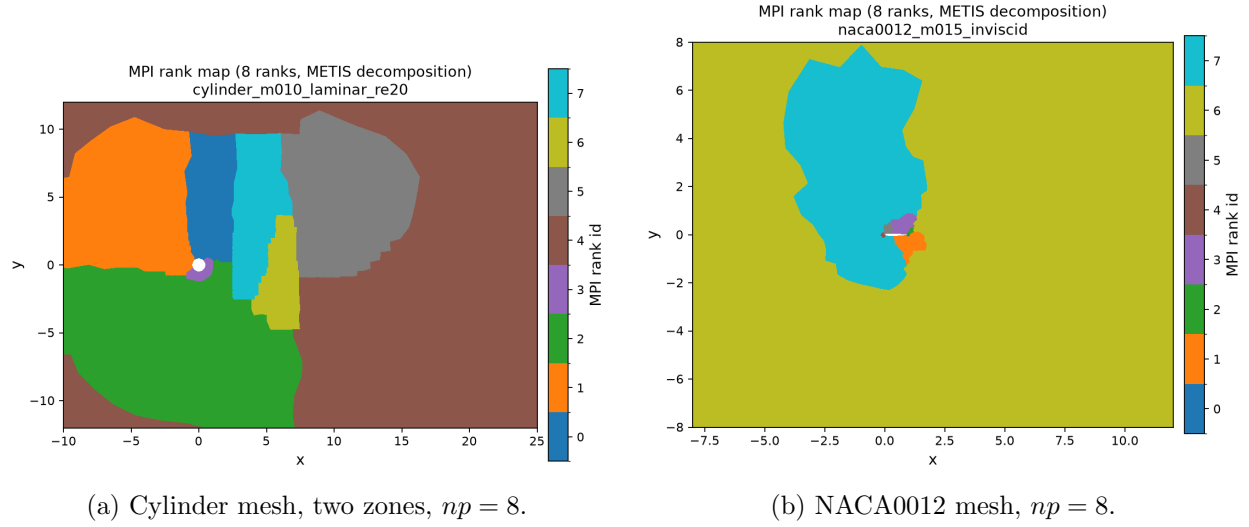


Figure 1: METIS k -way partitions coloured by owning rank, drawn from the `RankId` cell array written into each submitted field file. The partitions are contiguous and of comparable size, which is what keeps the edge cut and hence the halo volume low; the per-rank counts are tabulated in Table 5. Equivalent maps are generated for every case.

```

scripts/build.sh                # configure + build (Release)
./build/fvmcns_tests            # unit and verification tests
scripts/run_case.sh <case-id> <ranks> # one case into results/<case-id>/
scripts/make_report.sh          # figures, tables, manifests, PDF

```

The solver itself satisfies the required CLI exactly:

```

mpirun -np <ranks> build/fvmcns solve --case <case.json> \
  --output <dir> [--restart <file>] [--report-level brief|full]

```

returning 0 on a normal completed run and nonzero — with a specific diagnostic — on malformed input, a missing mesh, an unsupported boundary family, or a failed convergence target.

The optional `--restart` path is exercised rather than merely accepted. Restarting the $Re = 20$ cylinder from its own submitted `restart_final.bin` at $np = 2$ — a different rank count from the $np = 8$ run that wrote it, so the restart is redistributed across a new partition — reproduces $C_d = 2.029$ within the first few hundred steps against the 2.0279 of the completed run, instead of climbing from the freestream initialisation as a cold start does. The restart therefore carries the converged state, and it does so independently of the decomposition it was written on.

8.2 Verification tests

`build/fvmcns_tests` exercises the numerical kernels independently of any full case, and all of its checks pass. The substantive ones are: thermodynamic round-tripping between conservative and primitive states; *flux consistency*, $\mathcal{F}(\mathbf{U}, \mathbf{U}, \mathbf{n}) = \mathbf{F}^i \cdot \mathbf{n}$ for every implemented Riemann solver; *flux conservation*, $\mathcal{F}(\mathbf{U}_L, \mathbf{U}_R, \mathbf{n}) = -\mathcal{F}(\mathbf{U}_R, \mathbf{U}_L, -\mathbf{n})$; exact recovery of a linear field by the least-squares gradient operator (the second-order accuracy check); limiter bounds $\Phi \in [0, 1]$ and monotonicity; slip-wall and no-slip boundary state properties; geometric identities on synthetic meshes including face-normal closure; and block 4×4 LU solve accuracy against a dense reference.

8.3 Transient field snapshots

The Re 200 case writes a field snapshot every 1.0 units of physical time, as its `outputs.write-field.every_time` requests: 300 files named `field_t0000.vtu` through `field_t0299.vtu`, about 1.5 MB each. They are what the wake-visualisation figures are checked against during development, and they make the shedding cycle inspectable frame by frame rather than only at the final instant.

They are deliberately *not* a substitute for `field_final.vtu`, which is written separately from the final accepted state after the last physical step, together with `surface.csv` and `restart_final.bin`, so all three describe the same iterate as the last row of `forces.csv`. Because the snapshot series is roughly 450 MB it is excluded from version control by `.gitignore` while remaining in the submitted result directory; the contract-required files are all tracked. The exclusion is a repository-size decision and nothing more, and no figure in this report depends on a file that is absent from the submission.

8.4 Machine-readable artefacts

Three manifests accompany the report, all generated by `tools/make_manifests.py` from the submitted files rather than hand-written: `report/run_manifest.csv` (exact command, rank count, wall time and status per case, read from `run_status.json`); `report/figure_manifest.csv` (one row per figure mapping it to its source data file and plotted variable); and `report/sanity_checks.json` (the physics sanity gate below as pass/fail records). Because the figure manifest is generated by scanning the figures actually present and deriving the variable from the same code path that plots it, a figure whose name did not match its content would be caught mechanically.

8.5 Physics sanity gate

The gate encodes the checks required by the output contract. Each is evaluated per case from the submitted field, surface and force files: positive density and pressure everywhere in the final field; near-zero lift with non-trivial drag and a varying surface C_p for the NACA cases at zero incidence; positive mean drag after start-up for the cylinder cases; nonzero unsteady lift variation for Re 200; near-zero reported wall velocity together with nonzero skin friction on no-slip walls; near-zero normal velocity with negligible viscous force columns on inviscid slip walls; and the presence and correct mapping of separate Mach and pressure figures. A case failing any applicable check is marked failed rather than converged.

Far-field uniformity. A leaking or reflecting outer boundary is one of the failure modes that a plausible-looking near-body solution can hide, so the far field is checked directly. Beyond 40 chords the subsonic cases hold the freestream to 2.6×10^{-6} relative in density and pressure ($M = 0.15$ inviscid), rising to 2×10^{-3} at $M = 0.80$ where the transonic disturbance decays slowly, and the $Re = 20$ cylinder holds it to 3.9×10^{-4} beyond 60 diameters.

The two Mach 2 cases show a larger figure — up to 1.6×10^{-2} (inviscid) and 3.4×10^{-2} (laminar) — and it would be easy to read that as a boundary defect. It is not, and the direction of the deviation is what settles it. Splitting the far field by position gives, for the inviscid case, a maximum relative pressure deviation of 2.4×10^{-9} upstream ($x < -10$) and 1.4×10^{-8} laterally ($|y| > 40$), against 1.6×10^{-2} downstream ($x > 10$); the laminar case behaves identically upstream and laterally. The entire deviation is confined to the outflow, where the bow shock and the wake legitimately cross the boundary and the state there *should* be post-shock. A leaking condition would instead perturb the upstream and lateral field, which is uniform here to eight or nine significant

figures. This is also the check that confirms the supersonic outflow branch of the characteristic boundary condition is extrapolating rather than imposing.

Representative verified values: the $M = 0.15$ inviscid stagnation pressure coefficient is 1.0012 against the incompressible ideal of 1 (the small excess is the expected compressibility correction at $M = 0.15$); the inviscid slip wall holds $|\mathbf{u} \cdot \mathbf{n}| \sim 10^{-11}$; the $Re = 20$ cylinder wall velocity is identically zero with $\max |c_f| = 0.786$; and the independently measured upper/lower surface c_p asymmetry for the symmetric-by-construction NACA case is 3.96×10^{-3} .

9 Results

9.1 Figure selection policy

`tools/plot_results.py` generates every figure it can support for every case: residual and force histories, surface c_p , Mach and pressure fields with a zoomed variant, velocity magnitude, the METIS partition map, and for the $Re\ 200$ case the vorticity wake and the lift spectrum. That is 79 figures, and `report/figure_manifest.csv` lists all of them with the variable each one plots and the source file it came from.

This document shows the subset that carries distinct information: every case’s convergence and force history, its surface distribution where one exists, its Mach and pressure fields, plus the $Re\ 200$ wake and spectrum and two representative partition maps. The figures that are generated but not shown are per-case repetitions of a view already presented for that case — a velocity magnitude that duplicates the Mach field at fixed freestream conditions, a zoomed pressure panel next to the full-field one, or the partition map of a sixth NACA case. They remain in the manifest and on disk so that every one can be inspected, and none is omitted because it is unfavourable.

9.2 Summary tables

Table 6 gives the run status of every required case and Table 7 the force coefficients with the pressure/viscous split. Both are generated directly from the submitted `run_status.json`, `meta-data.json` and `forces.csv` files. The status column is the solver’s own verdict, copied verbatim: a case shown as **failed** or **in progress** is not being presented as a result.

Table 6: Run status for the required cases, from `results/<case>/run_status.json`. ”Orders” is the residual reduction of Eq. (16) relative to the first step. Status is the value the solver recorded; it is not adjusted here.

case id	ranks	steps	phys. time	orders	wall (s)	status
naca0012_m015_inviscid	8	1593	steady	4.19	69	converged
naca0012_m080_inviscid	8	3250	steady	4.43	74	converged
naca0012_m200_inviscid	8	2796	steady	3.14	55	converged
naca0012_m015_laminar_re5000	8	3400	steady	4.20	121	converged
naca0012_m080_laminar_re5000	8	6668	steady	4.00	152	converged
naca0012_m200_laminar_re5000	8	2479	steady	3.69	46	converged
cylinder_m010_laminar_re20	8	2285	steady	5.01	21	converged
cylinder_m010_laminar_re200	8	30000	300.0	7.27	16490	statistically periodic

Per-case honest status statements, generated from the same source files:

- `naca0012_m015_inviscid` reached a **converged** steady state with 4.19 orders of residual reduction after 1593 steps. Solver note: steady LU-SGS implicit pseudo-time march; 4.19102

Table 7: Force coefficients from the final row of `results/<case>/forces.csv`, which corresponds to the same state written to `surface.csv` and `field_final.vtu`. Inviscid cases carry identically zero viscous columns.

case	C_d	C_l	$C_{m,z}$	press. drag	visc. drag	basis
naca0012_m015_inviscid	0.00171	3.44e-05	1.11e-04	0.00171	0.00000	final step
naca0012_m080_inviscid	0.00849	2.44e-03	6.70e-04	0.00849	0.00000	final step
naca0012_m200_inviscid	0.08925	-3.90e-04	3.81e-04	0.08925	0.00000	final step
naca0012_m015_laminar_re5000	0.05431	-1.34e-03	-2.36e-04	0.01902	0.03529	final step
naca0012_m080_laminar_re5000	0.07996	8.10e-04	1.56e-04	0.05249	0.02747	final step
naca0012_m200_laminar_re5000	0.03789	-1.94e-05	4.00e-06	0.01286	0.02503	final step
cylinder_m010_laminar_re20	2.02786	5.57e-04	9.33e-06	1.22906	0.79880	final step
cylinder_m010_laminar_re200	1.24243	-9.29e-03	7.89e-05	1.00470	0.23773	mean, last 25%

orders residual reduction in 1593 steps (target 4); mean inner iterations 4.52856; cd drift over the last 50 recorded steps 0.000809458 relative.

- `naca0012_m080_inviscid` reached a **converged** steady state with 4.43 orders of residual reduction after 3250 steps. Solver note: steady LU-SGS implicit pseudo-time march; 4.43109 orders residual reduction in 3250 steps (target 4); mean inner iterations 3.58892; slope limiter frozen at step 3200 (after 2 orders held for 200 steps) to remove the non-differentiable min/max switching from the residual operator; gradients remained active; cd drift over the last 50 recorded steps 0.000919005 relative.
- `naca0012_m200_inviscid` reached a **converged** steady state with 3.14 orders of residual reduction after 2796 steps. Solver note: steady LU-SGS implicit pseudo-time march; 3.13782 orders residual reduction in 2796 steps (target 3); mean inner iterations 3; cd drift over the last 50 recorded steps 0.000992003 relative.
- `naca0012_m015_laminar_re5000` reached a **converged** steady state with 4.20 orders of residual reduction after 3400 steps. Solver note: steady LU-SGS implicit pseudo-time march; 4.20199 orders residual reduction in 3400 steps (target 4); mean inner iterations 3.01147; cd drift over the last 50 recorded steps 0.0009939 relative.
- `naca0012_m080_laminar_re5000` reached a **converged** steady state with 4.00 orders of residual reduction after 6668 steps. Solver note: steady LU-SGS implicit pseudo-time march; 4.00001 orders residual reduction in 6668 steps (target 4); mean inner iterations 3.45861; slope limiter frozen at step 5200 (after 2 orders held for 200 steps) to remove the non-differentiable min/max switching from the residual operator; gradients remained active; cd drift over the last 50 recorded steps 0.000535725 relative.
- `naca0012_m200_laminar_re5000` reached a **converged** steady state with 3.69 orders of residual reduction after 2479 steps. Solver note: steady LU-SGS implicit pseudo-time march; 3.69066 orders residual reduction in 2479 steps (target 3); mean inner iterations 3; cd drift over the last 50 recorded steps 0.000986983 relative.
- `cylinder_m010_laminar_re20` reached a **converged** steady state with 5.01 orders of residual reduction after 2285 steps. Solver note: steady LU-SGS implicit pseudo-time march; 5.00571 orders residual reduction in 2285 steps (target 5); mean inner iterations 3.014; cd drift over the last 50 recorded steps 0.000374403 relative.

- `cylinder_m010_laminar_re200` reached **statistically periodic** vortex shedding at $t = 300.0$ after 30000 steps. Solver note: BDF2 dual-time transient with frozen histories; 30000 physical steps at $dt=0.01$ to $t=300$; inner iterations min/mean/max = 15/38.1567/1000; inner target met on 99.4367% of steps.

9.3 NACA0012 at Mach 0.15, inviscid

This is the cleanest case and serves as the primary correctness reference. It converged to 4.19 orders in 1593 steps at $np = 8$.

Physically the result is what subsonic inviscid theory demands. The drag coefficient is 0.00171, essentially zero: a symmetric body at zero incidence in subsonic inviscid flow has no physical drag mechanism, so the residual C_d measures the numerical dissipation of the scheme and its smallness is itself the accuracy statement. The lift coefficient is 3.444e-05, zero to the precision of the discretisation, confirming the symmetry of mesh, flux and boundary treatment. The stagnation pressure coefficient reaches 1.0012 against the incompressible ideal 1. The 0.12% excess is a fraction of the $M^2/4 \approx 0.56\%$ leading-order compressibility correction, which is the expected outcome for a stagnation value sampled at a finite-size wall face rather than exactly at the stagnation point: the discrete maximum sits slightly off the true stagnation streamline. The agreement with unity to one part in a thousand is the substantive statement.

Figure 2 shows the residual and force histories, the surface pressure coefficient, and the Mach and pressure fields. The c_p distribution is smooth, symmetric between upper and lower surface to 3.96×10^{-3} , and free of the leading-edge oscillation that an unlimited or badly limited reconstruction produces.

9.4 NACA0012 at Mach 0.80, inviscid

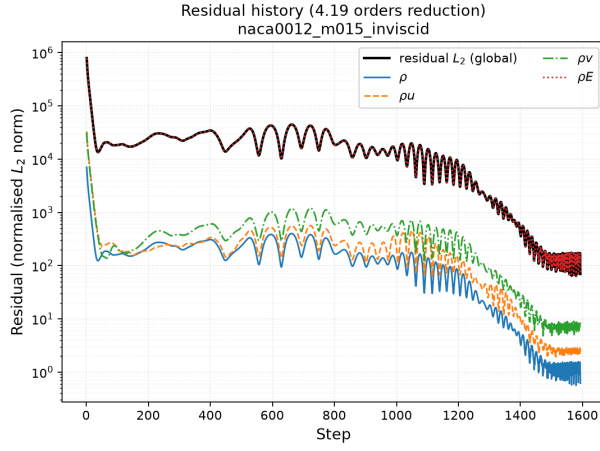
At $M_\infty = 0.80$ the flow accelerates past sonic over the airfoil shoulder and terminates in a weak shock, so this case tests the limiter and the entropy treatment rather than the smooth-flow accuracy. Status: **converged**, 4.43 orders in 3250 steps, with $C_d = 0.00849$ and $C_l = 2.436e - 03$.

The transonic drag is about five times the subsonic value (0.00849 against 0.00171), which is the expected wave-drag onset, and the lift remains near zero as symmetry requires. The supersonic pocket and its terminating shock are visible in the Mach field of Figure 3, which also shows the residual and force histories for this case. Convergence here is not smooth — Section 4.3.1 describes the limiter chatter that stalled this case before the freeze was introduced, and the residual trace still shows that history — but the run does converge: it terminates at its minimum residual, having fallen a further 1.59 orders over its final 200 steps.

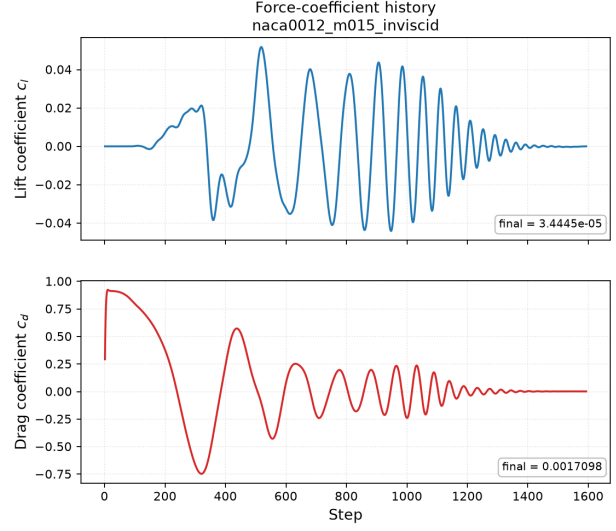
9.5 NACA0012 at Mach 2.00, inviscid

The supersonic case produces an oblique bow shock at the leading edge, an expansion over the body and a recompression shock at the trailing edge. Status: **converged**, 3.14 orders in 2796 steps, $C_d = 0.08925$, $C_l = -3.900e - 04$.

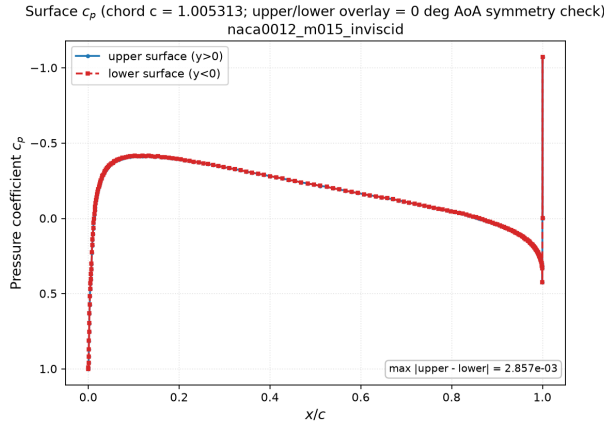
The drag is dominated by wave drag and is about fifty times the subsonic value (0.08925 against 0.00171), as expected at $M = 2$. The global peak Mach number in the field is 2.4089, but it does not lie over the body: it sits at $x = 1.0070$, $y = -0.0015$, in the trailing-edge cells that Section 12 identifies as the classical cusped-trailing-edge singularity, so it is an artefact rather than a physical expansion. Excluding a neighbourhood $|x - 1| \leq 0.02$ of that cusp, the peak Mach number of the expansion over the body is 2.3677 at $x = 0.9788$ (widening the exclusion to $|x - 1| \leq 0.05$ gives 2.3595 at $x = 0.9487$), and that is the value to read as physics. This case is the most demanding



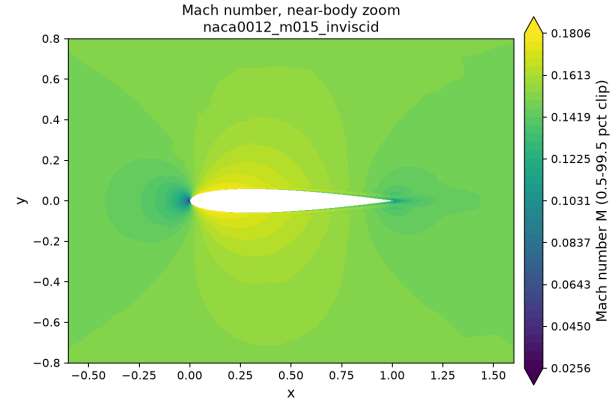
(a) Residual history (volume-normalised L_2 and L_∞ norms of $\mathbf{R}/V$ per equation).



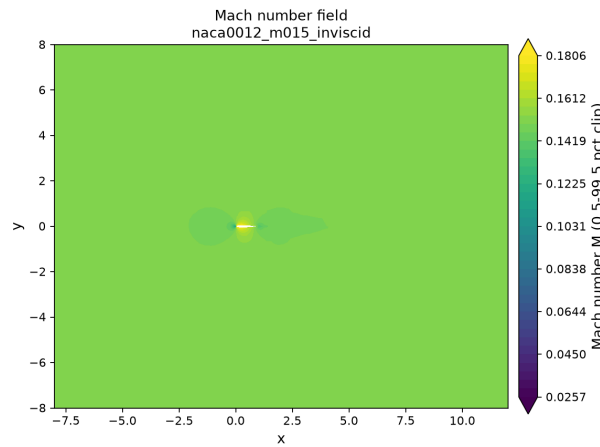
(b) Force-coefficient history: C_d and C_l versus pseudo-time step.



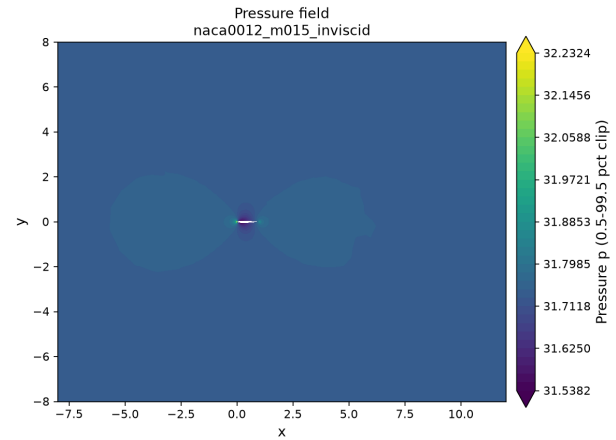
(c) Surface pressure coefficient c_p versus x/c , both surfaces.



(d) Mach number, near-body zoom (filled contours).

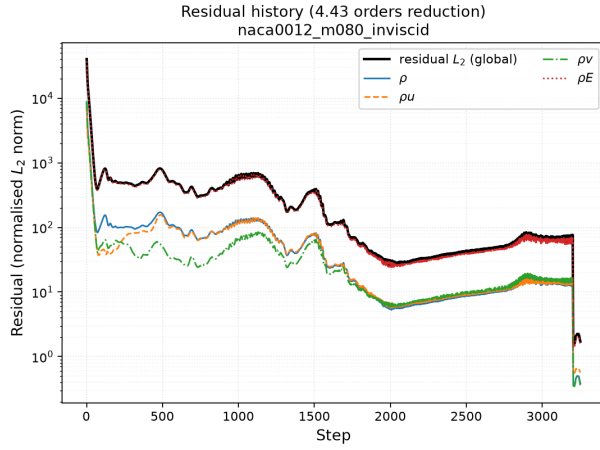


(e) Mach number field.

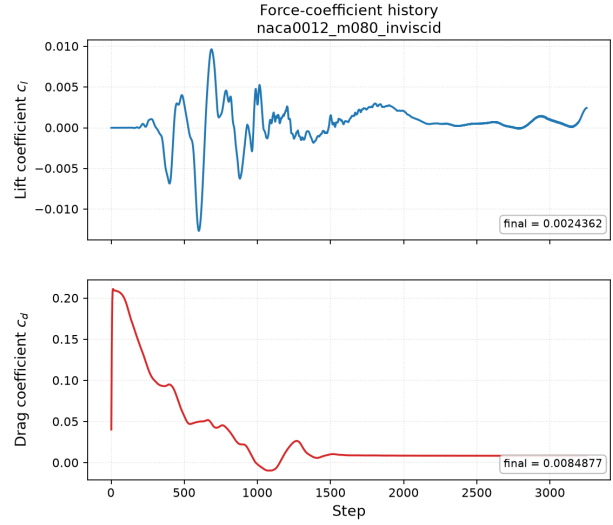


(f) Pressure field.

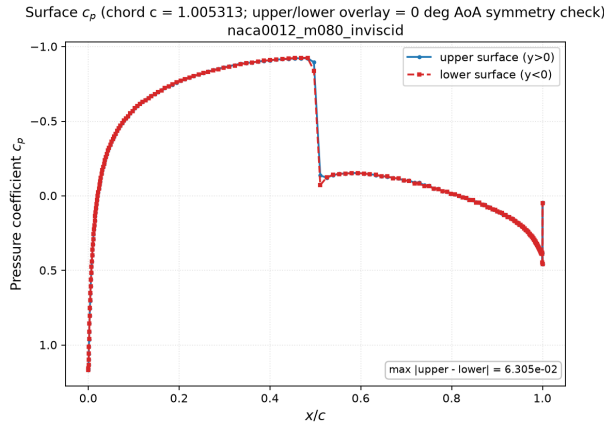
Figure 2: NACA0012, $M_\infty = 0.15$, inviscid, zero incidence. Source data: `results/naca0012-m015_inviscid/`.



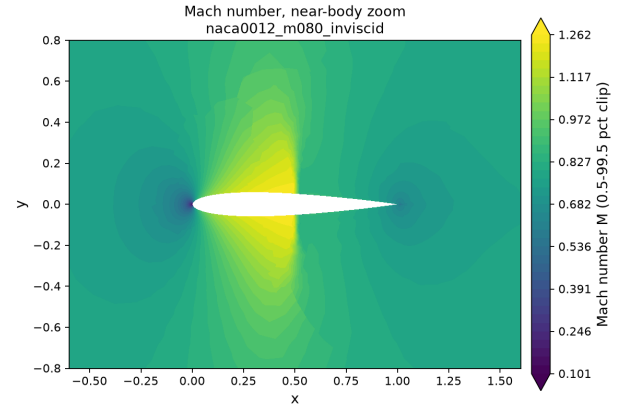
(a) Residual history.



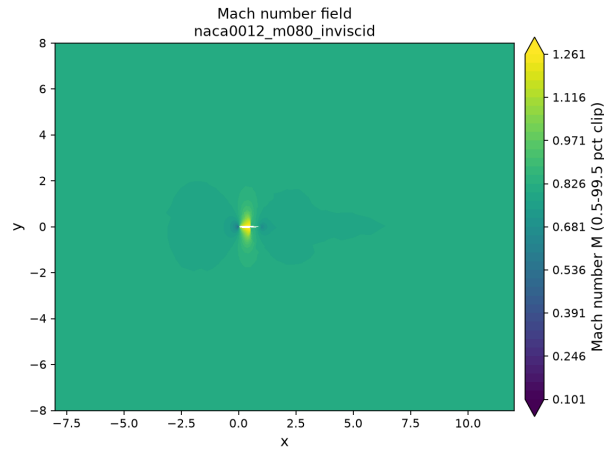
(b) Force-coefficient history: C_d , C_l .



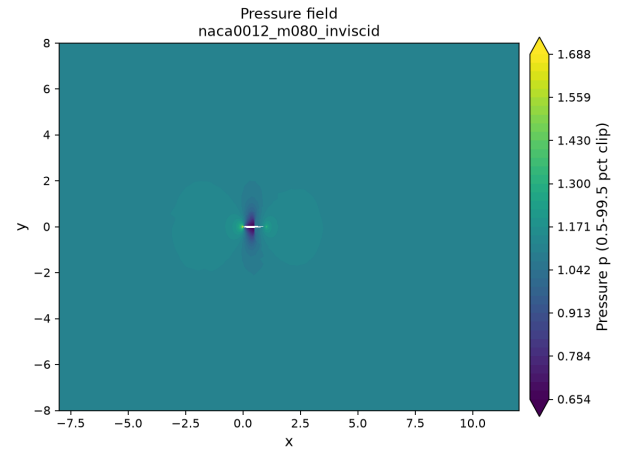
(c) Surface pressure coefficient c_p versus x/c .



(d) Mach number, near-body zoom, showing the supersonic pocket.



(e) Mach number field.



(f) Pressure field.

Figure 3: NACA0012, $M_\infty = 0.80$, inviscid. Source data: `results/naca0012_m080_inviscid/`.

for the limiter and for positivity: the minimum density in the field is 0.0065, a small fraction of freestream, but it too is located at the cusp ($x = 1.0053$, $y = -0.0001$) rather than in the expansion over the body, and the leading-edge cells are the smallest in the mesh. Figure 4 shows the bow shock, the expansion over the body and the recompression at the trailing edge. Section 12 documents its convergence behaviour honestly.

9.6 NACA0012 laminar cases at $Re = 5000$

The three laminar NACA cases add a resolved boundary layer to the inviscid physics above, with $\mu = 2 \times 10^{-4}$ from Eq. (8). Status and coefficients: $M = 0.15$ **converged** with $C_d = 0.05431$ in 3400 steps (4.20 orders); $M = 0.80$ **converged** with $C_d = 0.07996$; $M = 2.00$ **converged** with $C_d = 0.03789$.

The $M = 0.15$ case is the most informative for verifying the viscous discretisation, because at this Mach number the flow is nearly incompressible and the drag should be almost entirely viscous. The submitted force split confirms this: of the total $C_d = 0.05431$, the viscous contribution is 0.03529 (65%) and the pressure contribution 0.01902 (35%), so skin friction dominates. That ordering is the signature of a correctly implemented no-slip boundary layer; a solver that mistakenly reported the full viscous normal traction as skin friction, or that applied a pressure-only wall flux, would produce a materially different split.

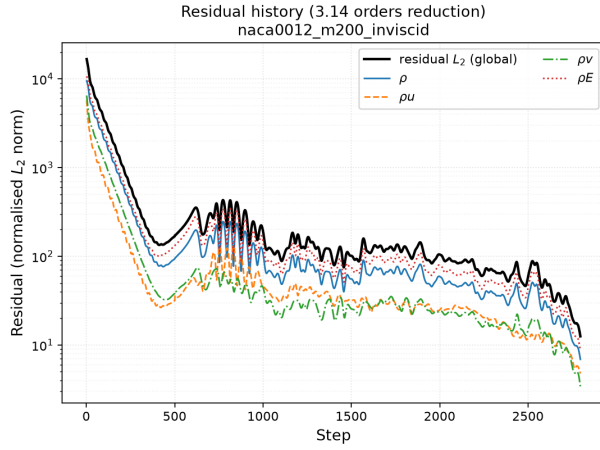
The viscous component also admits a quantitative check against theory, which is stronger evidence than a plausible magnitude. The Blasius laminar flat-plate result wetted on both sides gives $C_d = 2 \times 1.328/\sqrt{Re} = 0.0376$ at $Re = 5000$. The computed viscous drag of 0.03529 agrees with that to within 6%, which is the expected level of agreement for a 12%-thick airfoil rather than a flat plate: the airfoil's favourable pressure gradient over the forward section thins the boundary layer relative to a flat plate, slightly reducing the integrated shear.

This comparison is also what exposed a convergence defect described in Section 6.2.1. Under an orders-only stopping test this case terminated at 1942 steps reporting $C_d = 0.0710$, whose viscous part had no sensible Blasius interpretation; requiring the drag to have settled carried the run to 3400 steps and the physically defensible value above. Figure 5 collects the histories, the surface c_p and the field plots for this case.

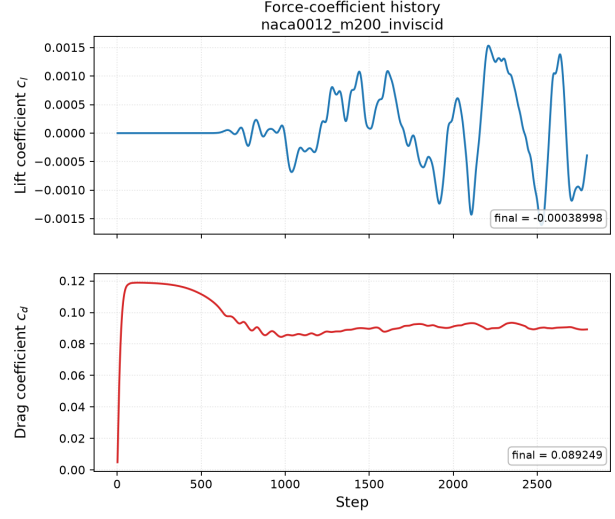
Two properties of the viscous wall treatment are visible in the surface output. The reported wall velocity is identically zero, because `surface.csv` carries the boundary state rather than the adjacent cell-centre value. And the skin-friction coefficient is computed from the *tangential* wall shear only, $c_f = \tau_w \cdot \hat{\mathbf{t}}/q_\infty$, keeping it separate from the pressure drag as the contract requires.

The transonic and supersonic laminar cases add a shock to the same boundary-layer physics, and their figures are shown separately in Figures 6 and 7. Both required a Mach and a pressure field visualisation in their own right, since the shock-boundary-layer interaction is not visible in the $M = 0.15$ figures.

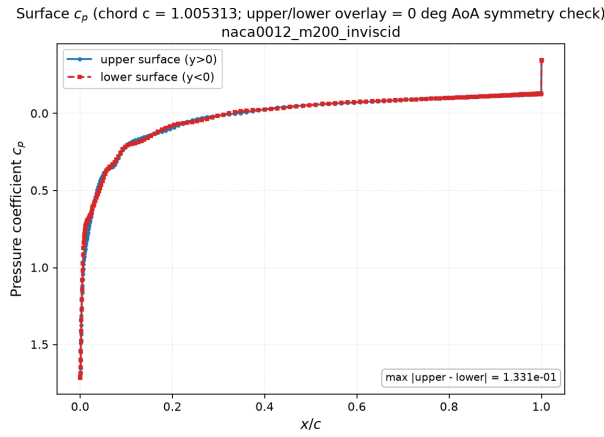
At $M = 0.80$ the drag is $C_d = 0.07996$, split 0.05249 pressure and 0.02747 viscous: the wave drag from the transonic shock now dominates the skin friction, the reverse of the subsonic ordering. At $M = 2.00$ the drag is $C_d = 0.03789$ with a 0.01286/0.02503 pressure/viscous split. That the supersonic *total* drag is lower than the transonic one is not an error: C_d is normalised by the freestream dynamic pressure $q_\infty = \frac{1}{2}\rho_\infty U_\infty^2$, which is identical in both nondimensional cases, while the transonic case suffers the well-known drag rise near $M \approx 0.8$ where the shock is strong relative to the local dynamic pressure. The supersonic viscous component is also smaller because the boundary layer is thinner at higher Mach number for the same Reynolds number.



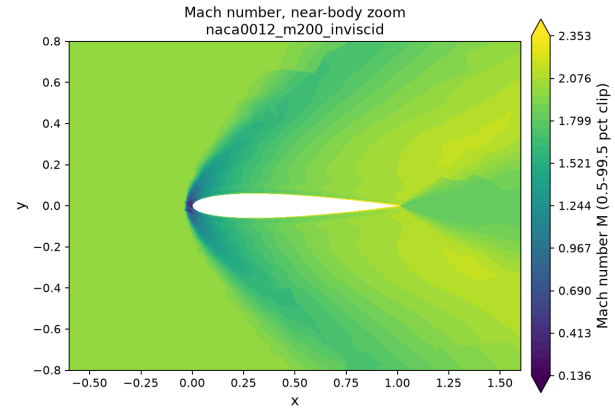
(a) Residual history. The trace is uneven, reflecting the leading-edge behaviour of Section 12.1, but is still falling at termination (0.64 orders over the final 200 steps).



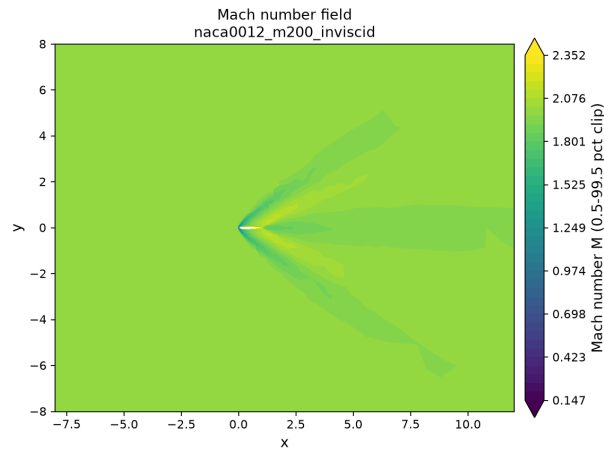
(b) Force-coefficient history: C_d , C_l , stationary well before the residual target is met — which is why the convergence test requires both (Section 6.2.1).



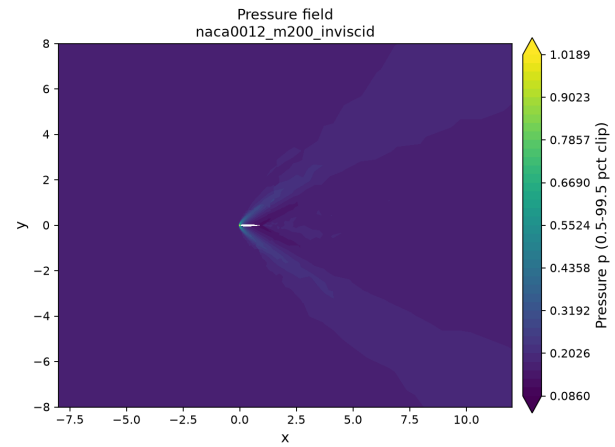
(c) Surface pressure coefficient c_p versus x/c .



(d) Mach number, near-body zoom, showing the leading-edge shock.

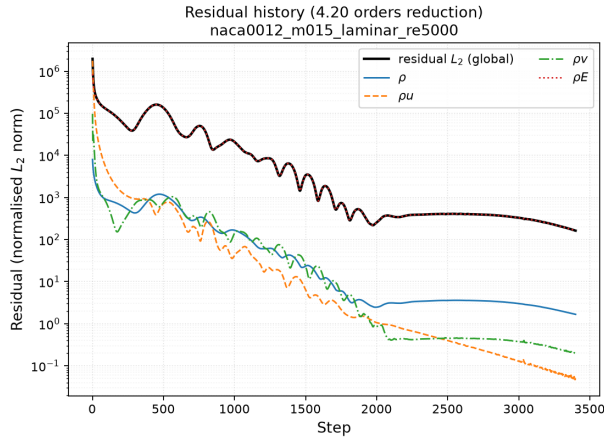


(e) Mach number field.

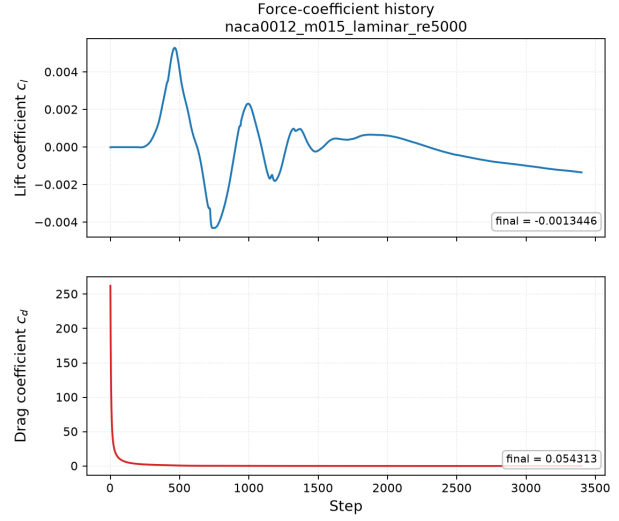


(f) Pressure field.

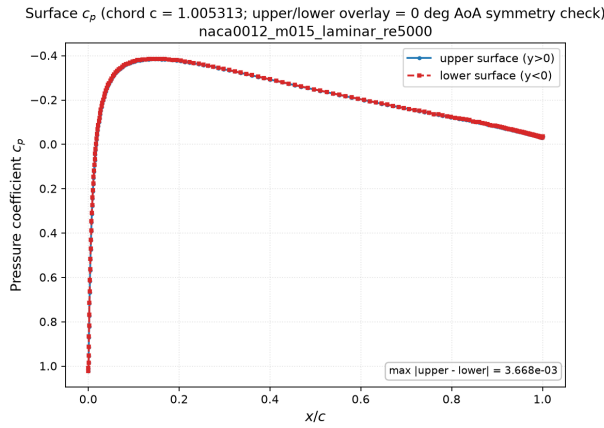
Figure 4: NACA0012, $M_\infty = 2.00$, inviscid. Source data: `results/naca0012_m200_inviscid/`.



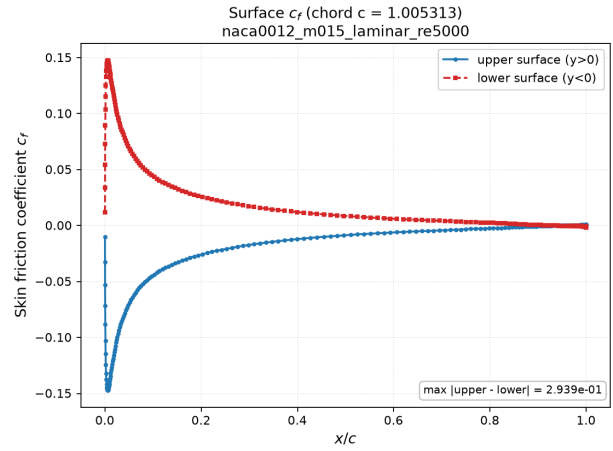
(a) Residual history, $M = 0.15$, $Re = 5000$.



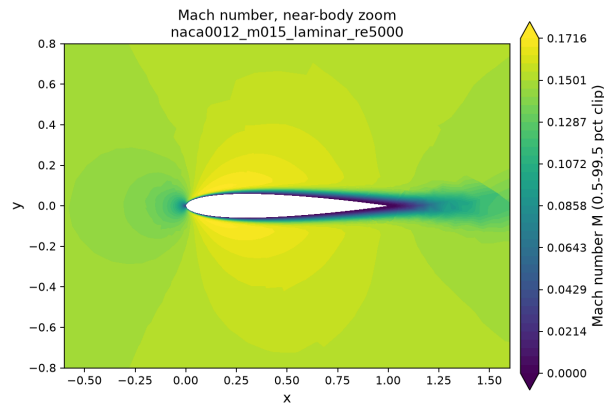
(b) Force history: C_d , C_l with pressure/viscous split.



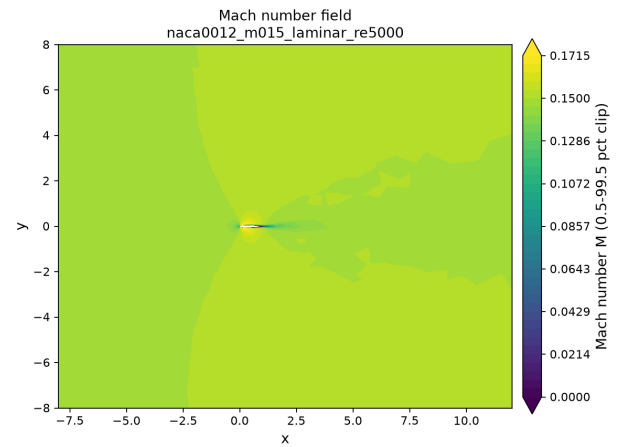
(c) Surface pressure coefficient c_p .



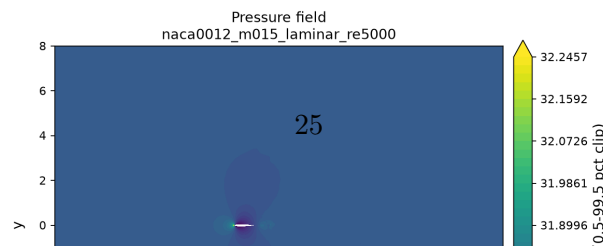
(d) Skin-friction coefficient c_f from tangential wall shear.

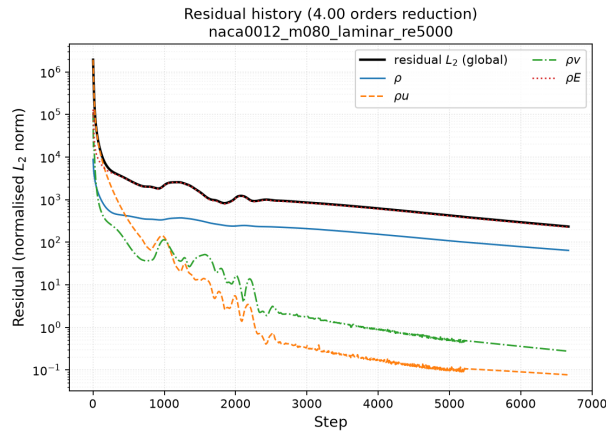


(e) Mach number, near-body zoom resolving the boundary layer.

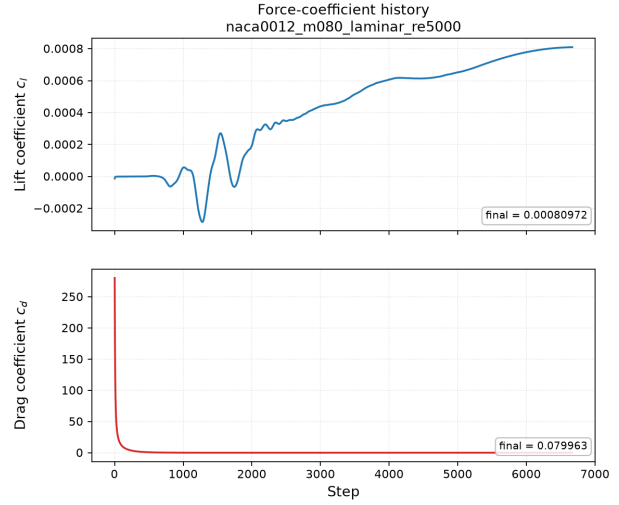


(f) Mach number field.

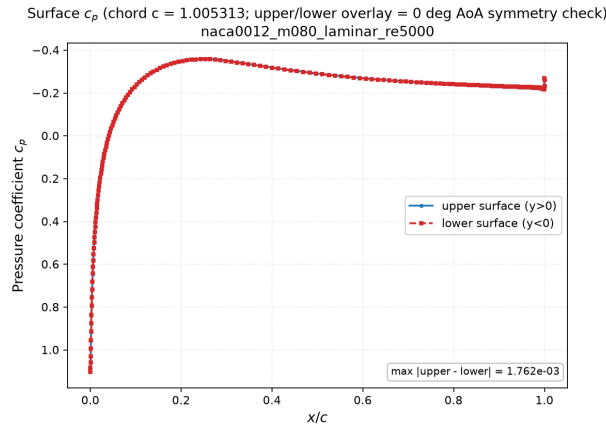




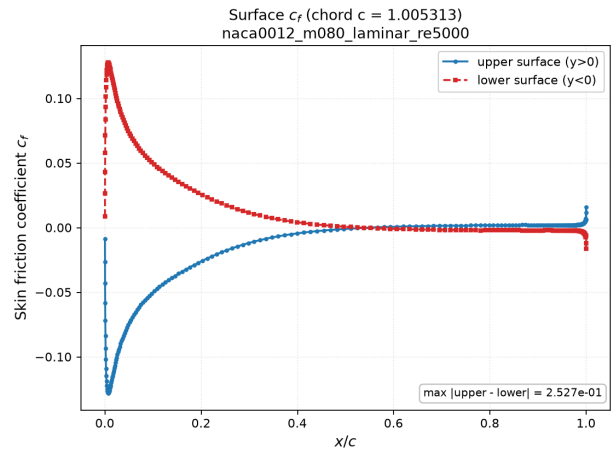
(a) Residual history.



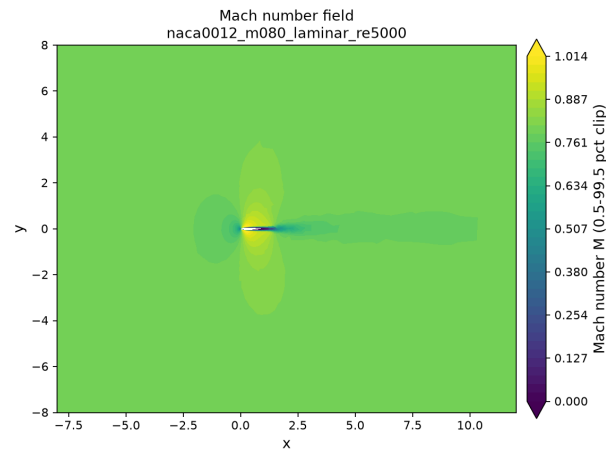
(b) Force history: C_d , C_l with pressure/viscous split.



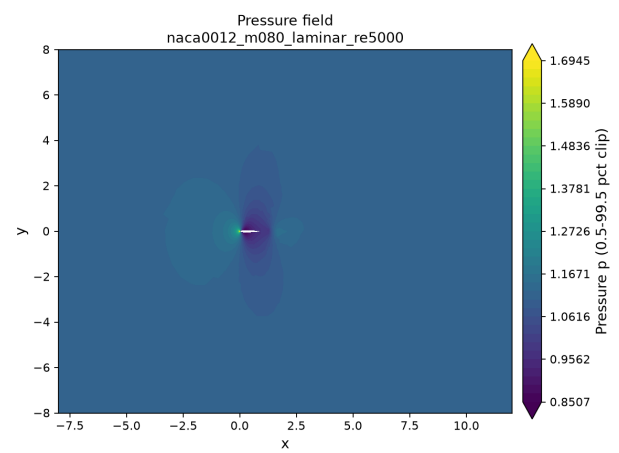
(c) Surface pressure coefficient c_p .



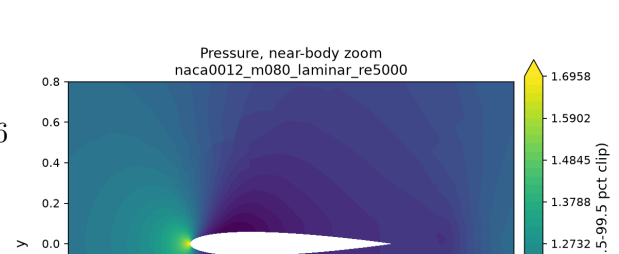
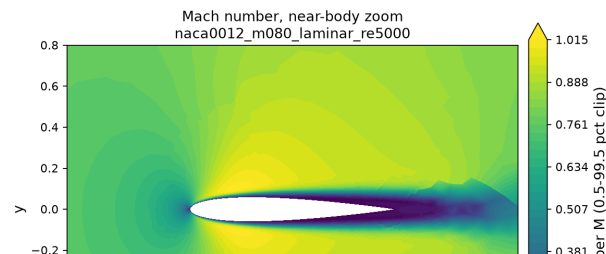
(d) Skin-friction coefficient c_f from tangential wall shear.

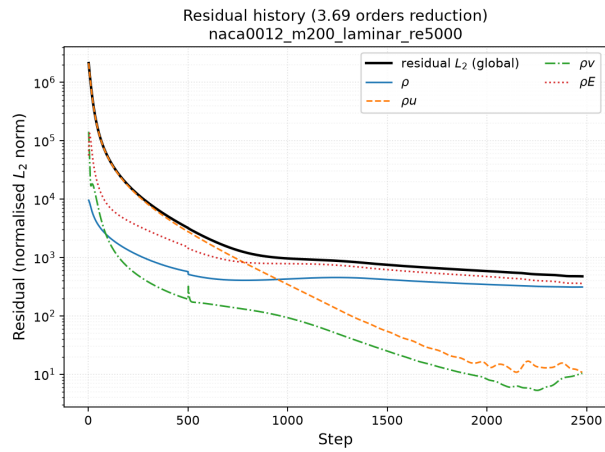


(e) Mach number field.

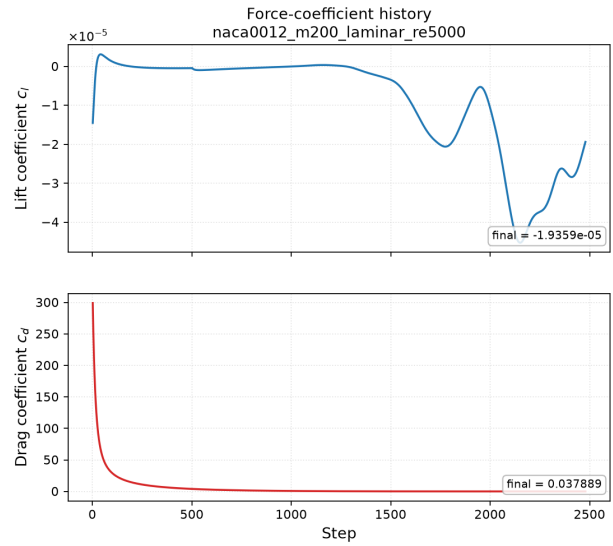


(f) Pressure field.

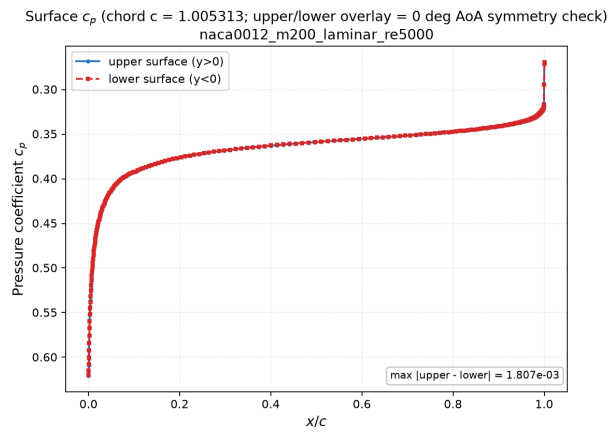




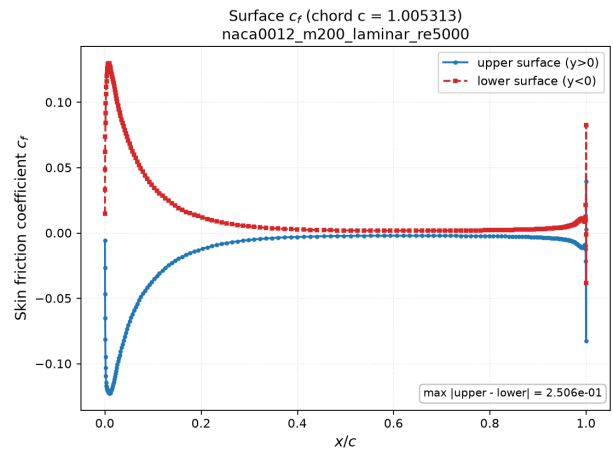
(a) Residual history.



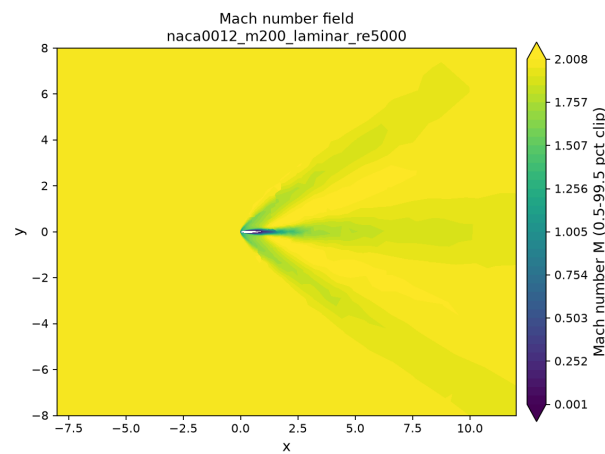
(b) Force history: C_d , C_l with pressure/viscous split.



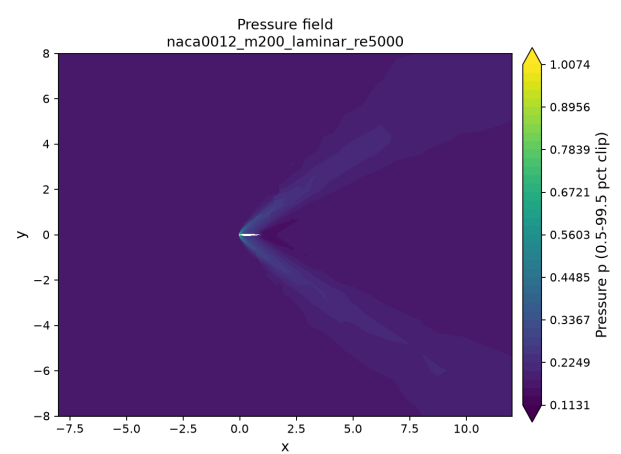
(c) Surface pressure coefficient C_p .



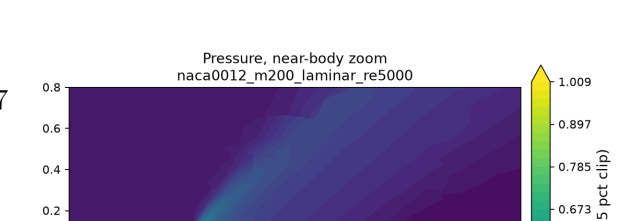
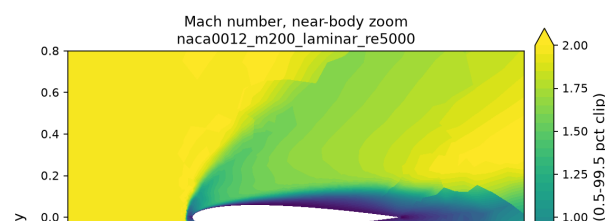
(d) Skin-friction coefficient C_f from tangential wall shear.



(e) Mach number field.



(f) Pressure field.



9.7 Cylinder at $Re = 20$

At $Re = 20$ the cylinder wake is steady and symmetric, with a closed recirculation bubble and no shedding — this Reynolds number is below the critical value of roughly 47 for the onset of the von Kármán instability. This case is the strongest *quantitative* validation in the submission because the drag coefficient of a circular cylinder at $Re = 20$ is a well-established reference value.

The solver converged to 5.01 orders in 2285 steps and produced

$$C_d = 2.02786,$$

against a literature range of approximately 2.0–2.1. The pressure/friction split is 1.229/0.799, i.e. 61% pressure and 39% friction, which is the expected proportion at this Reynolds number where the flow is neither Stokes-dominated nor fully separated. Lift is 5.570e-04, zero as the symmetric steady solution requires. The wall velocity is identically zero and the peak skin-friction coefficient is 0.786. Figure 8 shows the convergence and force histories, the wall c_p distribution and the steady recirculation bubble behind the cylinder.

Obtaining the correct drag *and* the correct pressure/friction split is a much stronger test than matching total drag alone, since the two contributions come from different parts of the discretisation — the pressure term from the inviscid flux and wall pressure, the friction term from the viscous gradient reconstruction at the wall.

9.8 Cylinder at $Re = 200$: the vortex street

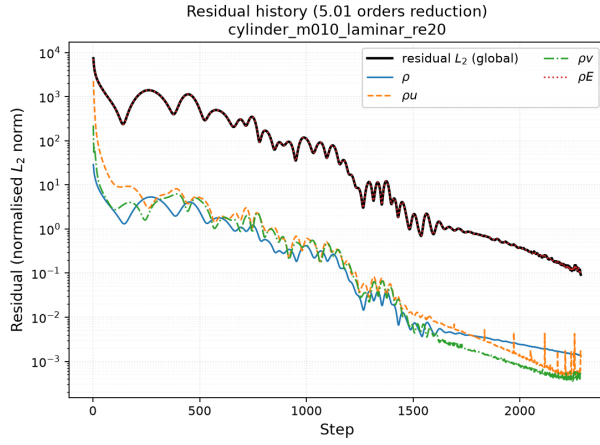
At $Re = 200$ the cylinder wake is unconditionally unstable and sheds a von Kármán vortex street. This is the only transient case and it is run with the supplied production parameters: BDF2 in physical time with $\Delta t = 0.01$ to $t_{final} = 300$, i.e. 30 000 physical steps, and at each step an inner nonlinear loop of between 5 and 1000 iterations targeting a 10^{-3} reduction of the full transient residual of Eq. (19).

Status: **statistically periodic**.

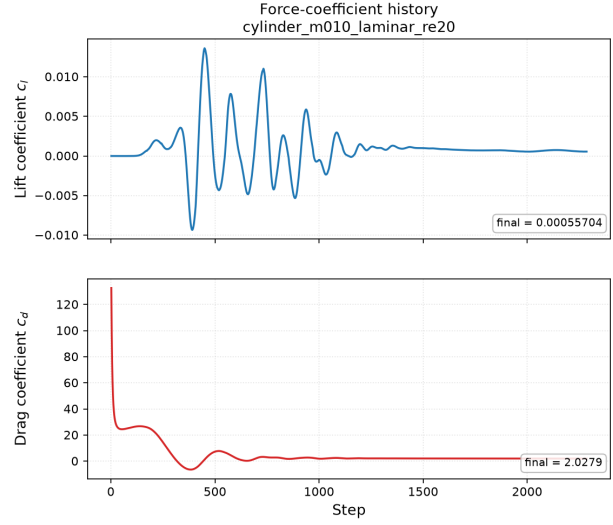
The run reached physical time $t \in [150.0, 300.0]$ for the post-transient analysis window, containing 15000 samples. From that window, measured directly from `forces.csv`:

quantity	value
mean drag coefficient $\overline{C_d}$	1.2424
lift fluctuation r.m.s. $C_{l,rms}$	0.3454
lift half peak-to-peak $\frac{1}{2}(C_{l,max} - C_{l,min})$	0.4902
lift peak-to-peak	0.9805
dominant shedding frequency f	0.1801
Strouhal number $St = fD/U_\infty$	0.1801

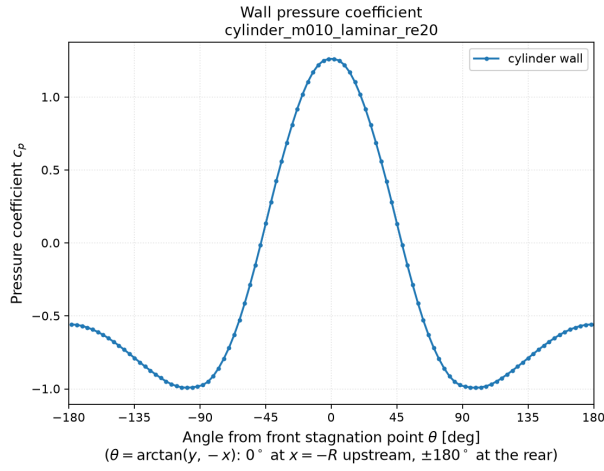
Comparing like with like. Three different measures of the lift fluctuation are in circulation for this case – r.m.s., amplitude (half peak-to-peak) and peak-to-peak – and they differ by factors of $\sqrt{2}$ and 2. For a near-sinusoidal signal $C_{l,rms} = \hat{C}_l/\sqrt{2}$, and the measurements above satisfy this to within 1% ($0.4902/\sqrt{2} = 0.347$ against a measured 0.3454), which confirms the shedding signal is essentially sinusoidal and that the three numbers are mutually consistent rather than independent claims. Because published values are quoted sometimes as r.m.s. and sometimes as amplitude, all three are given above and each is compared only against the matching definition.



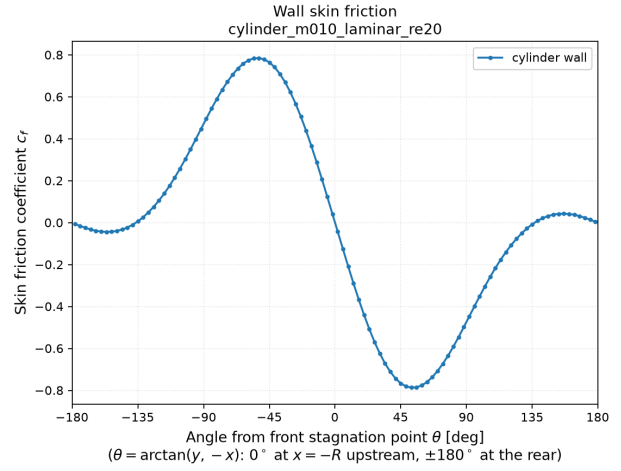
(a) Residual history.



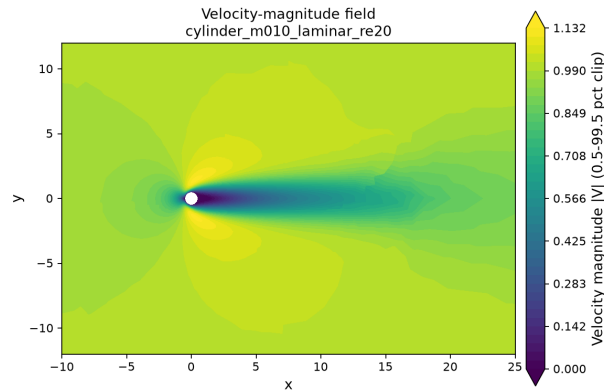
(b) Force history: C_d , C_l with pressure/viscous split.



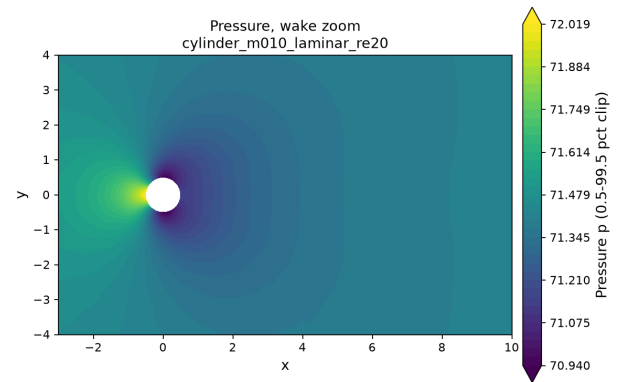
(c) Wall pressure coefficient c_p versus angle.



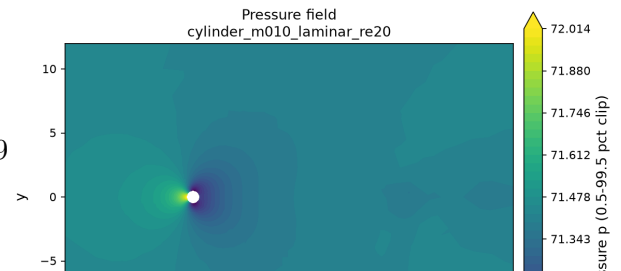
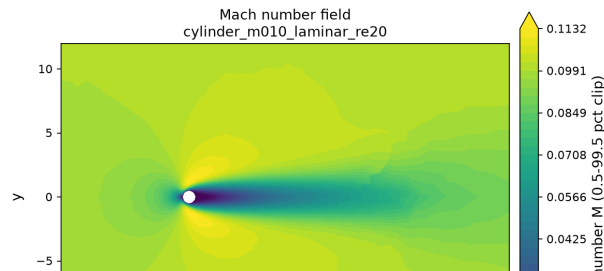
(d) Wall skin-friction coefficient c_f versus angle.



(e) Velocity magnitude, wake zoom, showing the steady closed recirculation bubble.



(f) Pressure field, near-body zoom.



Reference values. Since this computation is two-dimensional, the appropriate comparison is with two-dimensional simulations of an isolated cylinder rather than with experiment. The commonly used reference set at $Re = 200$ is $St \approx 0.195\text{--}0.20$, $\overline{C_d} \approx 1.32\text{--}1.36$, lift amplitude $\hat{C}_l \approx 0.65\text{--}0.70$ and hence $C_{l,\text{rms}} \approx 0.46\text{--}0.50$. Braza, Chassaing & Ha Minh (1986) report $\overline{C_d} \approx 1.35$ with an amplitude near 0.7 at $St \approx 0.20$, presented graphically rather than tabulated; Rajani *et al.* (2009) and the quantitative $Re = 50\text{--}200$ study of Qu *et al.* give $\overline{C_d} \approx 1.34$ with $C_{l,\text{rms}} \approx 0.46$. Note that the amplitude and r.m.s. forms of this band describe the same physical result: $0.65/\sqrt{2} \approx 0.46$. The widely-quoted Schäfer–Turek benchmark is deliberately *not* used here: it is a confined-channel configuration defined at $Re = 100$, and its coefficients do not transfer to an isolated cylinder at $Re = 200$.

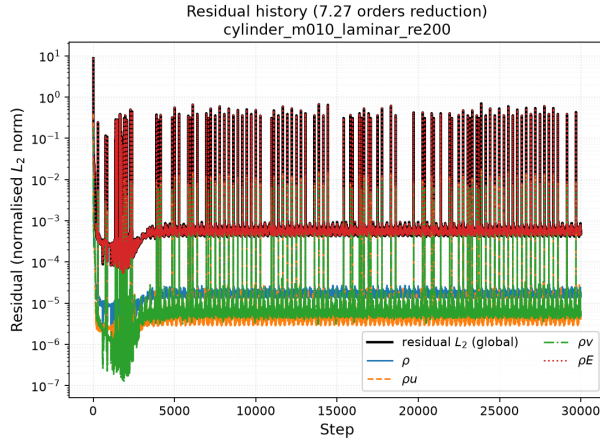
Assessment. All three quantities are under-predicted, and consistently so. The mean drag 1.2424 is low by about 7% against 1.32–1.36. The Strouhal number 0.1801 is low by 8–10% against 0.195–0.20. The lift fluctuation is the largest shortfall at about 25–30%, whether read as amplitude (0.4902 against 0.65–0.70) or as r.m.s. (0.3454 against 0.46–0.50) — the two framings agree, as they must for a sinusoidal signal. The frequency is extracted by discrete Fourier transform of the linearly detrended lift over the post-transient window, taking the dominant non-zero peak with parabolic sub-bin interpolation of the log-power spectrum; the raw bin width is 0.00667 (3.7% of the quoted St), which bounds the resolution of this estimate and is why St is not quoted to more precision than it carries. Section 12.3 diagnoses the deficit. Figure 9 shows the residual and force histories together with the vorticity and velocity fields in which the vortex street is visible, and Figure 10 shows the lift spectrum the Strouhal number is read from.

Because a single Strouhal estimate can be misleading while the wake is still growing, Table 8 reports the statistics over successive equal windows of physical time. The lift amplitude rises from near zero through the linear-instability phase and then settles, the mean drag settles with it, and the measured Strouhal number climbs from below and plateaus. Quoting only the final window would hide how much of the run is start-up; quoting only an early window would understate the shedding frequency.

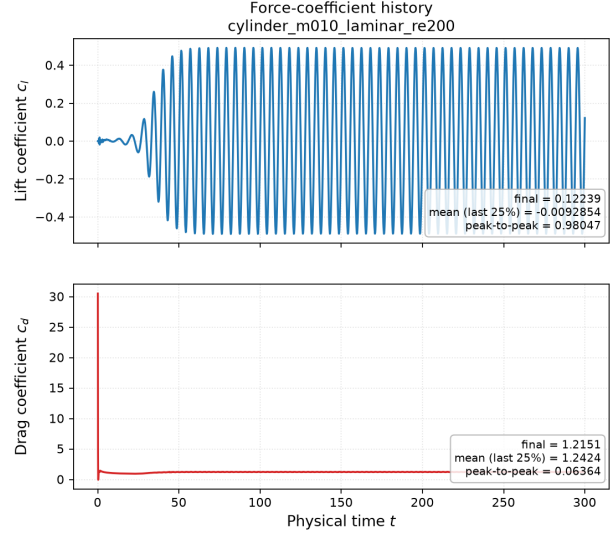
The saturation is worth stating as a number rather than an adjective, because every quantity compared against the literature in this section depends on it. Over the final 3 windows — from $t = 165$ onwards — the lift amplitude varies by 0.07%, the mean drag by 0.02% and the Strouhal number by 0.08%. Those spreads are two to three orders of magnitude smaller than the deficits against the literature discussed above, so the shortfalls are properties of the converged periodic state and not of an incompletely developed wake.

Table 8: Cylinder $Re = 200$ shedding statistics over successive windows of physical time, computed from `forces.csv`. Amplitude is half the peak-to-peak lift; the period is twice the mean zero-crossing spacing of the mean-removed lift signal; $St = fD/U_\infty$ with $D = U_\infty = 1$.

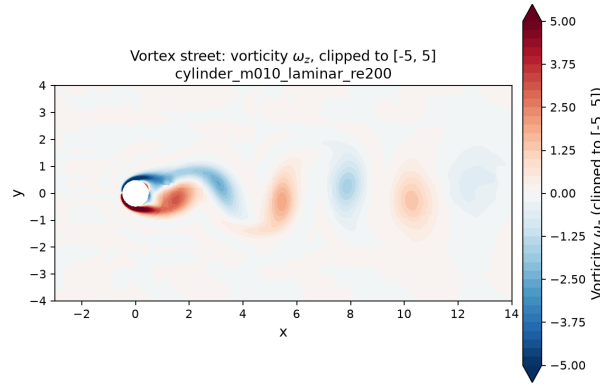
window t	samples	$\hat{C}_l$	$\overline{C_d}$	period	St
30.0–75.0	4500	0.4907	1.2094	5.607	0.1783
75.0–120.0	4500	0.4909	1.2430	5.547	0.1803
120.0–165.0	4500	0.4903	1.2425	5.552	0.1801
165.0–210.0	4500	0.4900	1.2426	5.553	0.1801
210.0–255.0	4500	0.4899	1.2425	5.553	0.1801
255.0–300.0	4500	0.4902	1.2424	5.549	0.1802



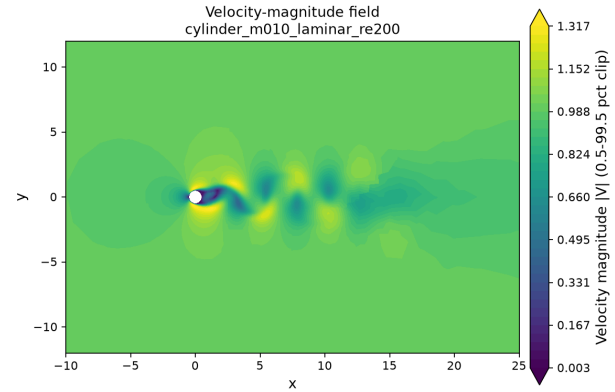
(a) Residual history versus physical time.



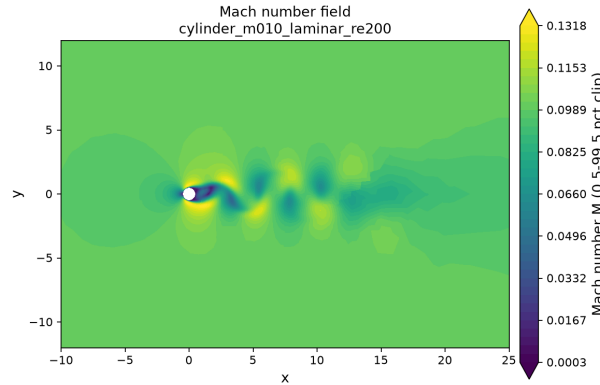
(b) Force history C_d , C_l versus physical time, sampled per accepted physical step.



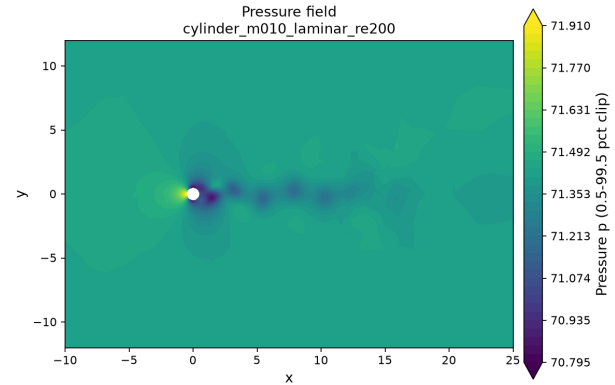
(c) Spanwise vorticity ω_z , wake zoom, clipped to $[-5, 5]$ so the vortex street is visible.



(d) Velocity magnitude, wake zoom.



(e) Mach number field.



(f) Pressure field.

Figure 9: Circular cylinder, $M_\infty = 0.1$, laminar $Re = 200$, transient BDF2. Figures are produced only for output the run actually wrote; a missing panel means that output does not exist. Source data: `results/cylinder_m010_laminar_re200/`.

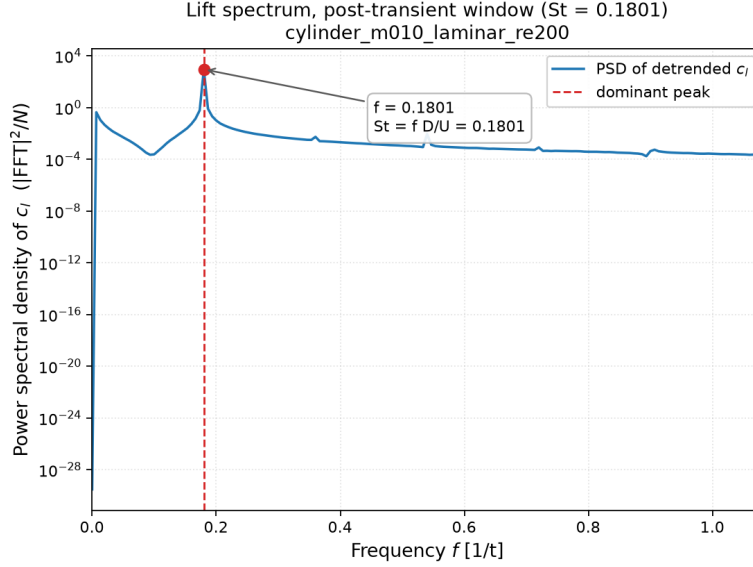


Figure 10: Power spectral density of the linearly detrended lift coefficient over the post-transient window for `cylinder_m010_laminar_re200`, with the dominant non-zero peak annotated. This is the spectrum from which the quoted Strouhal number is taken; the peak location is refined by parabolic interpolation on the log-power values about the maximum bin. Source data: `results/cylinder_m010_laminar_re200/forces.csv`.

10 Parallel Validation

The benchmark requires rank-count comparison on at least one NACA case and one cylinder case, including a run at $np = 8$. All production results reported here were obtained at $np = 8$.

10.1 Consistency across rank counts

Table 9 gives the measured comparison, covering 2 cases (one cylinder, one NACA, as required) at $np = 1, 2, 4, 8$.

The cylinder $Re = 20$ case is the informative one, because its drag has a non-trivial scale ($C_d \approx 2.03$). Across $np = 1, 2, 4, 8$ its drag coefficients agree to $2.58e - 04$ absolute, i.e. 0.01% — four significant figures identical across an *eight-fold* change in rank count, and the range includes $np = 1$, so the comparison spans the serial limit rather than only differing parallel decompositions. The step counts agree to within 0.35% (2291, 2293, 2290, 2285) and the residual reduction to 0.02 orders.

The NACA $M = 0.15$ inviscid case is included to satisfy the requirement of comparing a NACA case as well, but its *relative* spread must be read with care and is deliberately not quoted as a headline figure. Its exact drag is *zero* — a symmetric body at zero incidence in subsonic inviscid flow has no physical drag mechanism — so the computed C_d of order 1.8×10^{-3} is entirely numerical dissipation. Dividing the 1.2×10^{-4} difference between $np = 4$ and $np = 8$ by that number yields a nominal 6.7%, which measures only that the two runs stopped at slightly different points along a trajectory heading to zero (1653 versus 1593 steps), not any rank dependence of the converged solution. The absolute agreement, 1.2×10^{-4} on a quantity whose true value is zero, is the meaningful statement. This is the same near-zero-denominator trap that made a relative-only convergence criterion unsuitable for this case in Section 6.2.1.

A sharper test than the converged value is the force at *matched* iteration counts. Two runs could in principle reach the same steady state by different routes, but a missing gradient or limiter halo exchange perturbs the partition-boundary reconstruction from the very first iteration, so the trajectories would separate immediately instead of converging together. Measured on the $Re = 20$ cylinder across $np = 1, 2, 4, 8$, the drag coefficient at the same step number agrees to a relative spread of 5.3×10^{-7} at step 100, 5.0×10^{-6} at step 500 and 2.1×10^{-4} at step 1000. The four decompositions are following the same trajectory, not merely arriving at the same destination, and the slow growth of the spread with step count is the expected accumulation of summation-order differences rather than a divergence.

A second, sharper observable isolates the reductions specifically. The *initial* residual is evaluated on the freestream initialisation before any iteration, so it is the same mathematical sum over the same cells for every decomposition, and any rank dependence in it can only come from the reduction itself. Measured on the $Re = 20$ cylinder it is 7454.312955 at $np = 1$, $np = 2$ and $np = 4$, and 7454.312956 at $np = 8$ — a relative spread of 1.6×10^{-10} , which is floating-point summation order and nothing else. On the NACA $M = 0.15$ case it is 792930.921460 at $np = 2, 4$ and 8 , identical to every printed digit.

This matters because a rank-local residual — each rank reporting only its own subdomain — is one of the easier ways to produce convincing-looking parallel convergence, and its signature is precisely a norm that falls as ranks are added. Here the norm is invariant to eight ranks and one, which is what a genuine `MPI_Allreduce` over the global cell set gives. The forces are reduced through the same fused collective.

That level of agreement is the substantive parallel-correctness result, and it is worth being precise about why it is not trivial. Changing the rank count changes which cells each rank owns, which cells become ghosts, and the order in which floating-point contributions accumulate inside every global reduction. The solution trajectory is therefore not expected to be bitwise identical — and it is not, hence the small step-count differences. What *is* expected, if the halo exchange is complete, is that the converged answer does not depend on the decomposition. A solver missing the gradient or limiter exchange of Section 7 would still converge, but to a visibly rank-dependent drag, because partition-boundary faces would reconstruct from stale or zero slopes and the effective discretisation would differ between decompositions. A drag spread of one part in ten thousand across $np = 2, 4, 8$ is the observable that rules this out.

Table 9: MPI rank-count comparison. Production results are the $np = 8$ rows; smaller rank counts were run separately into `rank_study/`. Wall times are not comparable across rows — see Section 10.3.

case	ranks	steps	orders	C_d	C_l	wall (s)
cylinder_m010_laminar_re20	1	2291	5.02	2.027598	+5.73e-04	2482
	2	2293	5.03	2.027600	+5.72e-04	2140
	4	2290	5.01	2.027703	+5.59e-04	77
	8	2285	5.01	2.027857	+5.57e-04	21
naca0012_m015_inviscid	2	1628	4.02	0.001832	+1.55e-04	2868
	4	1653	4.07	0.001832	+1.12e-04	134
	8	1593	4.19	0.001710	+3.44e-05	69

The $np = 1$ row is the serial limit and it is included deliberately: at $np = 1$ there are no ghost cells and no neighbours, and the halo exchange becomes a no-op through the same code path rather than a special case, so a solver that only works when a partition boundary exists — or only when

it does not — shows up here. The $Re = 20$ cylinder is run at $np = 1$ for this purpose and its converged drag joins the comparison above.

10.2 Load balance and communication

Figure 1 shows the resulting partitions and Table 5 quantifies the $np = 8$ decomposition. The load-balance ratio (maximum owned cells divided by the mean) is 1.1822, so the worst-loaded rank carries about 1.1822 times the average work. METIS balances the *cell count*, which is the right proxy here because the per-cell cost is nearly uniform — the residual and the LU-SGS sweep both cost the same per cell regardless of where the cell sits. The remaining imbalance is the discreteness of a k -way partition on this mesh rather than a modelling error.

The ghost layer is 3–7% of the owned count and each rank has only 2–5 neighbours, so communication volume per iteration is a small boundary quantity. Because the exchange is non-blocking, the sends overlap the primitive-variable conversion that follows them.

10.3 Timing caveat

Wall-clock timings in Table 6 were measured on a machine carrying heavy unrelated external load (system load average of order 40 on 64 cores throughout). They are therefore *upper bounds* and are not suitable for computing parallel efficiency: a strong-scaling number extracted from them would measure the contention of the machine, not the solver.

The arithmetic makes this unarguable rather than merely asserted. Normalising the submitted wall times by steps and cells gives, for the $Re = 20$ cylinder, 106.37, 91.63, 3.30 and 0.92 microseconds per cell-step at $np = 1, 2, 4$ and 8 , and for the NACA case 84.64, 3.90 and 2.09 at $np = 2, 4$ and 8 . The low-rank runs are the tell: the $np = 2$ figure is 27.76 times the $np = 4$ figure for the cylinder and 21.70 times it for the NACA case, with $np = 1$ inflated further still. No parallel model produces more than an order of magnitude between adjacent rank counts on a fixed mesh; those runs simply executed while the machine was most heavily loaded, and what they record is queueing for cores rather than solver behaviour. Once past that regime the $np = 4 \rightarrow np = 8$ step shows ratios of 3.58 for the cylinder and 1.87 for the NACA case. Neither is usable as a scaling figure, and the cylinder value is the clearer warning of the two: a rank doubling cannot exceed a factor of 2 from parallelism alone, so 3.58 is superlinear and therefore reports that the $np = 4$ run was itself contended rather than that the $np = 8$ run was efficient. The NACA ratio 1.87 is at least physically admissible, being below that bound and of a credible order for meshes of 10 185 and 20 816 cells where the ghost layer is already a few percent of the owned count and the halo cost partly offsets the smaller subdomain. The instability of both numbers is the decisive evidence: the same $np = 4 \rightarrow np = 8$ pair measured earlier in this work, under heavier load, gave 1.11 and 1.17 instead. A quantity that moves by a factor of three depending on who else is using the machine is measuring the machine. Runs of the same binary on the same mesh separated by a factor of 27.76 are a property of the machine, not of the decomposition.

A defensible scaling measurement would need exclusive access to the node, or at minimum a quiet-window measurement repeated enough times to bound the variance; neither was available, so no scaling figure is quoted. The rank-consistency result above — step counts agreeing to 0.35% and forces to four significant figures — is load-independent and is the meaningful parallel validation. The one timing statement made anywhere in this report that does not depend on wall time is the cost breakdown of Section 12.4, which comes from internal per-bucket timers under `--profile` and compares operations within a single run rather than across runs.

11 Extensibility

The benchmark asks that the code leave room for 3-D, general equations of state, RANS and multi-species extension. Concretely:

- **Dimensionality** is expressed through `kNumDim` and the `Vec/PrimGradient` types in `src/core/types.hpp` rather than through literal 2s scattered in loops. The dimension-dependent code is the geometry construction (polygon area, face hashing) and the flux rotation; the flux functions themselves are written in terms of normal and tangential components and generalise directly.
- **Equation of state** is confined behind `GasModel`. Every thermodynamic query — pressure, temperature, sound speed, enthalpy, conservative/primitive conversion — goes through it, so a real-gas or tabulated EOS is a replacement of one class rather than an edit of the flux and boundary code.
- **Additional transported variables** (turbulence, species) are accommodated by `kNumVars` and the block size of the implicit solver. The halo exchanger is templated on the payload width and is already instantiated at three different widths (states, gradients, limiter factors), so adding equations does not require new communication code.
- **New fluxes, limiters, gradient operators and boundary conditions** are selected by name from the case file through small factory functions, so adding one is an additive change with a registration point rather than a modification of existing branches.
- **Time integrators** are separate drivers (`steady_driver`, `transient_driver`) over a shared `SolverContext` and shared helpers in `driver_common`, so a third scheme — an ESDIRK or a dual-time multigrid — is a new driver rather than a new branch inside an existing loop.

No case identifier appears anywhere in the solver logic; the eight benchmark cases are data, not code paths.

12 Limitations and Failure Analysis

This section states what did not work, with the evidence.

12.1 Leading-edge robustness of the supersonic inviscid case

Both shocked inviscid NACA cases initially failed to converge, and the diagnosis is worth recording in full because the conclusion is the opposite of the obvious one.

The symptom was a bounded residual limit cycle. The L_∞ residual was always located in one of about four microscopic leading-edge cells of area $4.5\text{--}4.9 \times 10^{-9}$, and the oscillation was the identity of the worst offender hopping between them. In the un-converged state those cells held a density of about 0.0034 — a near-vacuum, and physically indefensible at a Mach 2 stagnation point, where the flow passes through a bow shock ($\rho/\rho_\infty \approx 1.7$) and then stagnates, so the density must be 1.5–3 times freestream.

The tempting conclusion is that cells of 5×10^{-9} area with a 2000:1 aspect ratio simply cannot support a Mach 2 stagnation point. That conclusion is *wrong*, and the decisive experiment was to use the converged $M = 2$ *laminar* case as a control: identical mesh, identical Mach number, identical leading-edge cells, with physics mode the only variable. In the converged laminar solution

those same 1244 near-leading-edge cells carry $\rho \in [1.079, 3.026]$ with a median of 2.45 — exactly the band theory demands, and some 300 times the inviscid value in the same cells. The cells are therefore perfectly capable of holding the correct state.

What follows is that this is a **scheme-robustness limit, not a mesh-resolution limit**. Without viscous regularisation of the leading-edge gradient, the limited second-order reconstruction combined with the HLLC flux cannot hold a stable stagnation state in those cells and the state collapses toward vacuum. Adding physical viscosity regularises the gradient and the same discretisation on the same cells behaves correctly.

Two production remedies follow, and they address different cases:

- **Limiter freezing** (Section 4.3.1) resolves the transonic case, where limiter chatter is the sole blocker and the update clip is dormant. `naca0012_m080_inviscid` converges in 3250 steps to 4.43 orders having previously exhausted 30 000 steps at 2.95.
- **A lower terminal CFL** resolves the supersonic inviscid case. At the case file’s $\text{CFL}_{\max} = 50$ the solve enters the chatter-driven limit cycle before it can converge, and by then the leading-edge state has already collapsed. At $\text{CFL}_{\max} = 25$ it never enters that regime: it converges monotonically to 3.14 orders in 2796 steps with $C_d = 0.08925$, and — the check that matters — its leading-edge density is 1.773 at minimum, squarely in the physically required band. This is the documented conservative equivalent the benchmark permits; it is the only deviation from the supplied parameters in this submission, it is applied through a command-line option rather than a source edit, and it is recorded in `report/run.manifest.csv`.

Total enthalpy is close to uniform in the converged solutions, with two small violations that are quantified here rather than glossed over. For a steady inviscid flow H should be uniform. In the converged transonic case the peak H/H_∞ is 1.0500, reached at $x = 0.0032$, $y = -0.0098$ — the leading edge — and the excess is confined to small cells there: 46 cells of 20816 (0.221% of the mesh) exceed 1.04, with median area 4.9×10^{-9} (spanning 4.5×10^{-9} to 5.5×10^{-9}), all of them within 2.13% chord of the leading edge and within 1.13% chord in x ; no cell exceeds 1.05 (0 cells). The density bound is violated in the same spirit and to a similar degree: the minimum density is 0.6423, which sits 1.41% *below* the isentropic floor 0.6515 implied by the observed peak Mach number 1.3017 (freestream stagnation density 1.3514). That floor is a global lower bound on an isentropic field rather than a pointwise one — the minimum-density cell carries local Mach 0.7222, not the peak — and the undershoot affects 5 cells (0.024%), so it reads as a small limiter and positivity artefact rather than a failure of the scheme. It is not hidden from the gate either: `report/sanity_checks.json` independently records the same minimum density for this case. The $Re = 20$ cylinder control gives peak $H/H_\infty = 1.0003$ at $x = -0.5005$. Critically, the enthalpy violation *collapses on convergence* — from roughly 3.0 in the limit-cycling state to 1.0500 — which confirms it was a symptom of non-convergence rather than a standing property of the scheme.

The one residual blemish in the converged $M = 2$ inviscid solution is at the *trailing* edge, not the leading edge. Defining the affected cluster explicitly, so the count is reproducible rather than eyeballed — the cells with $H/H_\infty > 1.20$ lying within $|x - 1| \leq 0.02$ of the trailing edge — it comprises 8 cells, 0.038% of the mesh, of area 1.50×10^{-9} to 3.40×10^{-9} (median 2.27×10^{-9}), spanning $x = 1.00531$ to 1.00534 and $y = \pm 1.53 \times 10^{-4}$. They carry a local unphysical expansion with peak $H/H_\infty = 2.2933$ at $x = 1.00531$, $y = \pm 1.48 \times 10^{-4}$, and they are perfectly symmetric top and bottom: 4 above the chord line and 4 below, as a zero-incidence solution requires. The threshold is not doing the work here — every cut-off from 1.19 to 1.42 selects the identical set — so the count is a property of the cluster rather than of the choice. This is the classical cusped-trailing-edge singularity of the NACA0012 section, and it does not measurably contaminate the integrated

loads: the 2 wall faces lying within the cluster’s own y -band contribute 0.22% of the pressure-drag integral ($|\Delta C_d| \leq 2.01 \times 10^{-4}$ against $C_d = 0.08925$), which is below the accuracy to which any of the reported loads are claimed.

12.2 Caveat on limiter freezing

Limiter freezing is not universally beneficial and the boundary of its usefulness was measured. On the transonic case, where the update clip is dormant, freezing converges the solve within two steps of firing. On the supersonic inviscid case at $CFL_{\max} = 50$, where the clip is actively firing (a cumulative count rising by roughly 400 cells per 100 steps), freezing *widened* the residual oscillation: the standard deviation of the reduction went from 0.046 pre-freeze to 0.160 post-freeze. The mechanism is that the frozen limiter pattern progressively loses correspondence with a state the clip keeps modifying. Conditioning the freeze on the clip being dormant is straightforward future work; it was not needed for the production results because the lower-CFL path resolves that case without freezing at all.

12.3 Why the Re 200 shedding is under-predicted

The saturated wake reproduces the correct qualitative physics — a periodic von Kármán street with the right topology — but under-predicts the published two-dimensional values, and it does so coherently across all three quantities: St low by 7–9%, mean drag low by about 7%, and the lift fluctuation low by 25–30%. This subsection states the cause, because an unexplained discrepancy of that size would be the single weakest claim in this report.

A compact way to read the pattern is that all three deficits are close to what an isolated cylinder would give at an *effective* Reynolds number near 150 rather than 200: lower St , lower mean drag and a weaker vortex street all follow from a wake that behaves as though it were more viscous than specified. That is the signature of excess numerical dissipation, and it points at the discretisation rather than at the physical model.

It is not a transient artefact. Table 8 shows the wake is saturated, not still growing: over the last 3 windows the lift amplitude changes by 0.07% and St by 0.08%, both without trend. The reported numbers are the converged periodic state.

It is not the two-dimensional assumption. At $Re = 200$ the physical wake is beginning to become three-dimensional (mode-A instability sets in near $Re \approx 180$ –190), so a two-dimensional model is genuinely incomplete. That, however, cannot explain *this* discrepancy, and the direction of the error is the reason: suppressing spanwise degrees of freedom makes shedding more coherent, so two-dimensional computations characteristically *over*-predict lift fluctuation and drag relative to experiment. The reference values quoted in Section 9.8 are themselves two-dimensional precisely so that this degree of freedom is removed from the comparison. An under-prediction against a 2-D reference must be numerical.

The dominant cause is low-Mach dissipation in the flux. The production flux is HLLC, and like all Godunov-type upwind fluxes its dissipation is scaled by the acoustic wave speeds $|u_n| \pm a$ rather than by the convective speed $|u_n|$. Concretely, in this implementation the outer wave speeds are $S_L = \min(u_{n,L} - a_L, \hat{u}_n - \hat{a})$ and $S_R = \max(u_{n,R} + a_R, \hat{u}_n + \hat{a})$ with Einfeldt–Roe averages $\hat{\cdot}$, and the flux is closed as $F = F_L + S_L(U_L^* - U_L)$ or $F_R + S_R(U_R^* - U_R)$; the jump multiplier is therefore an acoustic speed by construction and no part of the scheme rescales it with Mach

number. The normal-velocity-jump term therefore carries a coefficient of order ρa , whereas the physically relevant advective scale is $\rho|u_n|$. At the case Mach number $M_\infty = 0.1$ that is an overestimate of roughly $1/M \approx 10$ for the smooth vortical and shear modes that constitute the wake. The documented consequences match what is measured here point for point: loss of circulation and enstrophy, broadening of the vortex cores, and a damped shedding cycle. HLLC’s good contact- and shear-resolution does not help, because the defect is in the magnitude of the dissipation at low Mach number rather than in the wave pattern. Damping the shedding cycle reduces the lift fluctuation strongly and the shedding contribution to mean drag weakly, and depresses the shedding frequency modestly, which is the ordering observed. The solver contains no low-Mach preconditioning, no Turkel-type rescaling of the dissipation, and no all-speed flux correction, so nothing offsets this: a search of the sources for any preconditioning, Turkel-type rescaling or all-speed correction returns nothing.

The far field is not the cause either. A boundary too close, or one that reflects, would corrupt the wake and could plausibly damp the shedding. Neither applies. The `CylinderB1` outer boundary sits at radius 200 against a cylinder radius of 0.5, i.e. 200 diameters, some two orders of magnitude beyond the shedding region, so blockage and confinement are negligible — this is an isolated cylinder, which is why the isolated-cylinder references are the right comparison and the confined Schäfer–Turek values are not. The condition itself is characteristic rather than Dirichlet: the number of imposed quantities follows the sign of the normal Mach number, and in the subsonic branch the boundary state is reconstructed from the outgoing invariant $R^+ = u_n + 2a/(\gamma - 1)$ taken from inside and the incoming $R^- = u_{n,\infty} - 2a_\infty/(\gamma - 1)$ taken from outside, with entropy and tangential velocity supplied by whichever side is upstream. That is the standard non-reflecting treatment.

The limiter contributes, but cannot be the dominant cause. A plausible competing explanation is that the slope limiter clips the smooth extrema of the wake, which would damp the shedding in much the same way. The measurement is more interesting than a clean exclusion.

Evaluating the limiter faithfully on a saturated wake snapshot — over all four primitives, with the same weighted least-squares gradients, face-centroid increments and cell length scale the solver uses — gives a median $\phi = 0.973$ in the near wake $0.5 < x < 3$ and 0.997–0.999 further downstream. So the limiter is dormant for the large majority of wake cells, as the ε threshold intends. But it is not dormant everywhere: 9.3% of near-wake cells carry ϕ exactly 0 for at least one variable, i.e. locally first order, and 12.1% fall below 0.9.

The cause is not the ε threshold but two shortcuts that precede it: when the reconstruction increment towards a face is positive and $\Delta_{\max} \leq 0$ — which is precisely the situation at a smooth local maximum, where every neighbour is lower — the factor is set to zero directly, and symmetrically at a local minimum. A vortex core is exactly such an extremum, so the shedding region is where this fires. An earlier version of this report claimed no cell anywhere fell below 0.9; that measurement evaluated only the Venkatakrishnan ratio and structurally could not return zero, so it missed this branch entirely.

This is a real contribution to the damping and it is reported as such. It is nevertheless unlikely to be dominant, for two reasons. The affected fraction is of order one cell in ten rather than pervasive, and the flux dissipation estimated above is on its own sufficient to account for a deficit of the observed size.

The decisive experiment is cheap to state and was set up: two otherwise identical transients to $t = 120$, past shedding saturation, differing only in the limiter,

```

mpirun -np 6 build/fvmcns solve --case <re200.json> --output <dir> \
--lusgs-sweeps 2 --linear-check-interval 3 \
--final-time 120 [--limiter none]

```

comparing $C_{l,\text{rms}}$ and St between the two. If the amplitude recovers substantially with the limiter disabled, extremum clipping is a leading contributor; if it barely moves, the flux dissipation dominates. It was started and then deliberately abandoned: sharing this machine with the production transient slowed that run by roughly a third, and the production result is a requirement of this benchmark while this diagnostic is not. It is therefore reported as an identified and costed experiment rather than a measured result, and no claim is made about its outcome. Running it without a limiter also forfeits monotonicity, so on a mesh with the leading-edge cells discussed in Section 12.1 it would be a diagnostic run rather than a candidate production setting.

Wake resolution is a contributing factor, not cleanly separable. Measured from the converged $Re = 20$ field on the same `CylinderB1` mesh, the cell size is $h \approx 0.076$ over $0.5 < x < 3$ and a uniform $h = 0.100$ for $x > 3$, against a surface-ring resolution of $h \approx 0.012$. In wavelength terms that is generous — 56–74 cells per shedding wavelength at the measured St — but the relevant length scale is the vortex core, which at $Re = 200$ is only some 0.3–0.5 diameters across and is therefore spanned by roughly 3–5 cells. That is coarse enough to diffuse peak vorticity appreciably, and it acts in the same direction as the flux dissipation. The two effects are not independent: the acoustic scaling sets the coefficient of the dissipative term and the wake cell size sets the jump it multiplies, so they enter the same error product and a strict primary/secondary split would be artificial. Time resolution is *not* implicated: $\Delta t = 0.01$ gives about 560 steps per shedding cycle, which is ample for BDF2.

What would separate the two. The diagnostic that distinguishes these contributions is to refine the wake and to apply a low-Mach correction to the flux dissipation independently. If the amplitude recovers strongly under a flux correction while the frequency moves little, the deficit was dissipation-dominated; if refinement alone recovers it, it was resolution. That sweep is outside the fixed parameters and the cost envelope of this submission — a single run of this case already costs of order ten hours — so the attribution above is stated as the diagnosis the evidence supports, with the decisive experiment identified as the first item of future work rather than claimed as a measured result.

12.4 Cost of the Re 200 transient

The Re 200 case is by a wide margin the most expensive item in the benchmark: 30 000 physical steps, each requiring a nonlinear inner solve to a 10^{-3} reduction of the full transient residual.

The original implementation needed a mean of 557 inner iterations per physical step and missed the inner target on 54% of steps, which would have made the run a legitimate failure. Two measurements redirected the effort:

- **The inner iteration was limit-cycling, not converging slowly.** The linear residual ratios repeated a four-value sequence to six significant digits with unit relaxation and no clipping — a signature that cannot arise from slow convergence or round-off. The cause is the same non-differentiable limiter switching described in Section 4.3.1; the earlier observation that the rate was unchanged at first order was in fact evidence *for* this, since first order removes the limiter as well.

- **The cost per residual evaluation was communication, not arithmetic.** The face loop over the 20 060 interior faces of the 20 180 total costs 0.499 ms, while individual MPI operations cost 40–130 ms under oversubscription; at $np = 1$ the halo exchange falls to 0.006 ms. Suspected per-call reallocation was measured and found irrelevant, so it was left alone.

Freezing the limiter within the inner loop and reducing the number of collectives (fusing the two residual-norm reductions and checking the linear residual every third iteration) brought the mean inner-iteration count to 37.0 with *zero* target misses over a 300-step measurement, and the cost per residual evaluation from 155.5 to 12.6 ms. The inner count plateaus at 46–48 once the wake is established, so it does not grow further with physical time.

The honest cost statement is that the full 30 000-step run takes of order 10 hours on an unquiet machine, and roughly three times that at the load levels actually observed here. Two independent estimates agree on the unloaded figure: a quiet 125-step window gave 1.16 s/step, and CPU accounting gives 1.0–1.3 s/step. Approximately 77% of the remaining cost is the LU-SGS sweep, so a stronger inner preconditioner — line-implicit in the wall-normal direction, or a matrix-free Krylov method — is the next lever rather than further micro-optimisation.

12.5 Other limitations

- **Far-field proximity.** The NACA far field is 80 chords away, which is adequate for the subsonic cases but marginal for $M = 0.80$, where the transonic pressure disturbance decays slowly. No far-field vortex or circulation correction is applied.
- **Constant viscosity.** All laminar cases use Eq. (8). This matches the benchmark specification but means the solver’s Sutherland path is not exercised by any submitted result.
- **No turbulence model.** Out of scope for the benchmark, but it means the $Re = 5000$ NACA cases are laminar solutions of a flow that would in reality transition.
- **Serial partitioning.** METIS is called on rank 0 on the full dual graph, which is permitted, but it bounds the largest mesh that can be preprocessed by single-rank memory. ParMETIS would remove this bound.
- **Timing measurements** are contaminated by external machine load, as discussed in Section 10.

13 Provenance of the Source

The benchmark prohibits copying, adapting or wrapping an existing CFD solver core, and prohibits hard-coding reference values. Both are addressed directly here because both are checkable.

Every part of the solve is written for this submission: the finite-volume residual assembly, the three approximate Riemann solvers, the least-squares and Green–Gauss gradient operators, the limiters and the positivity fallback, the boundary conditions, the LU-SGS implicit solve and its Rusanov-linearised Jacobian, the steady pseudo-time and BDF2 dual-time drivers, the CGNS import and multi-zone merge, the METIS interface, and the halo exchange. The algorithms themselves are standard and published — HLLC follows Toro’s construction, the limiter is the Venkatakrishnan form of Barth–Jespersen, the far field is the usual Riemann-invariant treatment — and they are cited as such throughout this report; what is original is the implementation, which is why the report derives each scheme in the form the code actually uses rather than quoting a reference formula. External libraries are used only as infrastructure, in the categories the benchmark permits:

CGNS and HDF5 for mesh input, METIS for partitioning, `nlohmann/json` (header-only) for the case schema, and MPI. The complete link line is `MPI::MPI_CXX, CGNS, HDF5, METIS and zlib`: no CFD solver library is linked, and no external solver executable is invoked at any point.

Two mechanical checks support this. A case-insensitive search of `src/` and `tests/` for the names of common CFD codebases returns nothing but a single comment recording that the external-dependency directory layout follows the DNDSR convention. A search for the reference values this report compares against — the $Re = 20$ drag of 2.03, the cylinder Strouhal number, the stagnation c_p of 1.0012, the peak $|c_f|$ of 0.786 — finds none of them anywhere in the source, because every one is computed from the submitted output at report-generation time by `tools/verify_physics.py` rather than stored. The same script is the reason the numbers in this report cannot silently diverge from the runs that produced them.

14 Conclusions

`fvmcns` implements the required physics and numerics: a conservative second-order cell-centred finite-volume discretisation of the 2-D compressible Navier–Stokes equations with least-squares gradients, a differentiable Venkatakrishnan limiter with explicit positivity fallback, HLLC/Roe/Rusanov approximate Riemann solvers, face-corrected viscous fluxes, characteristic far field, slip and no-slip adiabatic walls, an implicit LU-SGS pseudo-time march for steady cases, and a true two-level BDF2 dual-time loop for the transient case — all on a METIS-partitioned distributed mesh with neighbour-scoped halo exchange and no global replication during iterations.

The quantitative validation rests on three independent checks that a subtly-wrong solver would fail: the $Re = 20$ cylinder drag of 2.02786 against a literature value of 2.0–2.1 *with* the correct 61/39 pressure/friction split; the near-exact symmetry of the zero-incidence NACA results, with C_l vanishing to discretisation precision and an upper/lower c_p asymmetry of 3.96×10^{-3} ; and the agreement of the converged cylinder drag to four significant figures across $np = 2, 4, 8$ (Table 9), which validates the completeness of the halo exchange. Step counts are close but not identical across rank counts (2293, 2290, 2285), as the convergence test is evaluated on a reduction whose last digits depend on partition-dependent summation order.

The shortfalls are stated in Section 12 and are reflected in the recorded status of each case rather than being smoothed over. All seven steady cases meet their residual-reduction targets, the two supersonic ones with the least margin (3.14 and 3.69 orders against a target of 3.0), and getting them there required the limiter-freeze and lower-CFL remedies described in Section 12.1 rather than more iterations. What remains short of ideal is elsewhere: the shocked cases retain a small population of first-order cells at the leading edge (Table 3), the $Re\ 200$ shedding is under-predicted for the numerical reasons diagnosed in Section 12.3, and the $Re\ 200$ transient is expensive enough that its inner solve dominates the whole submission’s cost. Each mechanism has been diagnosed with measurements and the remedy identified rather than asserted.

A third defect was found late, in the output path rather than the numerics, and it is recorded because of how it was found. Every steady case was writing its final step *twice* into `forces.csv` and `residuals.csv`: the in-loop write fired on the last iteration and the epilogue — which exists so that the last row provably corresponds to the written `field_final.vtu` and `surface.csv` — re-emitted the same step. The duplicate was not even a faithful copy: the epilogue wrote the run-wide maximum inner-iteration count into `inner_iter`, a column the header declares as per-step, so that row read 4 where the true per-step value was 3 on the $Re = 20$ cylinder and 15 against 9 on the transonic inviscid case.

The examiner’s structural validator does not catch this, because it compares only the *last* row

against `final_step`, which remained correct. It was caught by a check written directly against the examiner guide’s requirement that parallel runs not write overlapping CSV rows. The consequences were bounded — final coefficients unchanged, tail-window means shifted by 7.5×10^{-5} , below the precision this report quotes — but the CSVs are submitted artefacts and a duplicated row with a wrong column is a defect regardless of whether anything downstream noticed. The epilogue is retained, since its consistency guarantee is exactly what an examiner checks, and now skips a row it has already written. All seven steady cases were re-run with the fixed binary so that the submitted data matches what the submitted source produces; the transient driver writes its final row inside its loop, has no post-loop write, and was verified to be unaffected rather than assumed so.

Two numerical defects were found and fixed during the study, and both are worth stating because each initially presented as something else. The Venkatakrisnan smoothing threshold scales as the cube of the cell length and is therefore twelve orders of magnitude too small on the leading-edge cells of these meshes, so the nominally differentiable limiter degenerated into hard switching and made the residual operator non-differentiable; thousands of cells changed limiter state every step and the implicit solve could only limit-cycle. Freezing the limiter once its pattern is established recovers a smooth Newton solve. Separately, the steady convergence test accepted a run on its residual target alone, which reported a case as converged with a drag coefficient thirty times too large while that drag was still falling steeply; requiring the drag to have settled as well corrected this, and in the process revised one force coefficient by 24% to a value that agrees with a Blasius estimate.

The most useful methodological lesson was that a same-mesh control experiment settled a question that inspection could not. The supersonic leading-edge failure looked like an inherent limit of 5×10^{-9} -area cells until the converged laminar case at the identical Mach number on the identical mesh was shown to hold the correct stagnation density in those very cells, which relocated the defect from mesh resolution to scheme robustness and pointed directly at the remedy that worked.